ATTACHMENT A

June 28, 2024 Ecora File No.: P240202

Town of Ladysmith
410 Esplanade – PO Box 220
Ladysmith, BC V9G 1A2

Attention: **Ryan Bouma, P.Eng., Director of Infrastructure Services | Town Ladysmith**

Reference: **Stocking Lake Detailed Design**

Ecora Engineering and Environmental Ltd. (Ecora) is pleased to submit this proposal to the Town of Ladysmith (ToL) in response to the requested services pertaining to Stocking Lake Dam Detailed Design. Ecora has assembled a strong team with extensive experience relevant to this assignment including completing dam safety reviews, dam engineering assessments, dam remediation design, hydrotechnical engineering, geotechnical engineering, and construction administering.

The price provided in this response will be guaranteed for a period of 60 days following the submission date. Please do not hesitate to call or email Michael J. Laws at 250.470.8808 or michael.laws@ecora.ca with any questions.

Sincerely

Ecora Engineering & Environmental Ltd.

Michael J. Laws, P.Eng.
Principal Dams & Geotechnics
Direct Line: 250.470.8808
michael.laws@ecora.ca



Stocking Lake Dam Remediation Detailed Design

Presented To:

Ryan Bouma, P.Eng., Director of Infrastructure Services

Town of Ladysmith
410 Esplanade – PO Box 220
Ladysmith, BC V9G 1A2

Dated: June 2024
Ecora File No.: P240202

Company Information: Ecora Engineering & Environmental Ltd.
200 – 2045 Enterprise Way, Kelowna, BC V1Y 9T5
Permit to Practice #: PP766

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1. Introduction

Ecora Engineering & Environmental Ltd. (Ecora) is pleased to submit this proposal to the Town of Ladysmith (ToL). We understand that the Town of Ladysmith is seeking professional engineering services to undertake detailed design for the remediation of Stocking Lake Dam.

Ecora is familiar with the Stocking Lake Dam as we have previously been engaged by the ToL to provide professional engineering services for the conceptual and preliminary design phases of this project in 2021 and 2022 respectively. Additionally, Ecora has provided emergency response design services in 2023.

Referencing the *Stocking Lake Dam Remediation Preliminary Design Report (Ecora, 2022)*, the proposed design includes the construction of a new dam downstream of the existing one in order to address deficiencies in the existing dam. The proposed replacement dam consists of:

- A vertical FRP (Fibreglass Reinforced Polymer) sheet pile centre wall (water retention element) with upstream and downstream rock embankments,
- A reinforced concrete spillway control drop structure and stilling basin,
- A 450 mm HDPE low-level outlet with upstream valve control, and
- A riprap lined spillway channel.

In addition to the above components, the design includes a plan for breaching the existing dam, an access road to the dam crest, a pedestrian trail to provide continuity with the existing trail network, and site grading to improve the stability of the spillway channel slopes.

The detailed design phase will include a collaborative effort between disciplines including Civil, Geotechnical, Hydrotechnical and Structural Engineering as well as environmental services. Ecora is pleased to offer all these services in house and will assign responsibilities for the technical components of the design upon completion, as per Engineers & Geoscientists British Columbia guidelines.

2. Proposed Scope of Services

Ecora proposes the following tasks to be performed as part of the Stocking Lake Detailed Design:

- Project Initiation and Kick-Off Meeting,
- Background Review,
- Site Reconnaissance,
- Hydrotechnical Analyses,
- Geotechnical Investigation & Analyses,
- Environmental Investigation,
- Detailed Design,
- Tendering/Request for Proposals,

- Construction Support Services,
- Environmental Monitoring, and
- Post-Construction Services.

2.1 Project Initiation and Kick-Off Meeting

Ecora's project manager, Adam Kerk-Hecker P.Eng., will organize a Kick-Off meeting with key Ecora personnel and ToL staff. During this meeting, introductions, lines of communication, project administration requirements including deliverable dates, quality control, and safety requirements will be established/confirmed. The meeting is anticipated to occur virtually to lower project costs.

2.2 Background Review

Ecora will carry out a desktop review of existing background information. Ecora's familiarity with this site and watershed from previous projects including the conceptual design, preliminary design, and emergency repairs of Stocking Lake Dam will expedite this process.

A LiDAR survey of Holland Lake and the surrounding area including Stocking Lake/Stocking Creek watershed is scheduled to be completed as part of the Holland Lake project Ecora is undertaking. This information can be used to supplement the detailed design of Stocking Lake dam. The LiDAR will also allow Ecora to undertake more accurate catchment assessment and inundation modelling.

Additional sources will be reviewed as necessary and aid in the review and completion of relevant hydrological and geotechnical investigation / analysis as well as the completion of a Construction Environmental Management Plan (CEMP) in line with the detailed design.

2.3 Site Reconnaissance

A site reconnaissance will be completed by the Professional of Record (PoR) in order to visually assess the conditions of the site. This will allow the team to confirm no major changes have occurred at the site since last visit and that the site investigation data collected as part of the conceptual and preliminary design remediation project is still relevant.

2.4 Hydrotechnical Analysis

The analysis completed during the conceptual and preliminary design of the dam will be assessed and any aspects that have not yet been completed, and are needed for the detailed designs would be completed in accordance with the following:

- Conduct a hydrological analysis to determine design flows, wind and wave effects and freeboard capacity of the proposed dam structure,
- Confirm hydraulic analysis of the low-level outlet, spillway and outlet channel using the US Army Corps of Engineers "HEC-RAS" modelling software. Hydraulic modelling will be completed to assess channel flow characteristics and design any necessary channel upgrades that may be required to mitigate against overtopping and erosion of the dam's hydraulic appurtenant structures, and

- With new LiDAR data becoming available as part of the Holland Lake dams project Ecora is also undertaking, dam breach analysis will be re-run with the new terrain data to confirm the inundation areas and proposed consequence classification as per the guidelines.

2.5 Geotechnical Site Investigation

2.5.1 General

In support of the Stocking Lake Dam detail design Ecora proposes a supplemental geotechnical site investigation. Existing geotechnical investigations have been undertaken by Ecora in addition to historical geotechnical investigations performed by other firms, however, these have been focused on the existing dam structure as summarized in section 2.5.3 below. This geotechnical site investigation would supplement the known data with information localized to the proposed new design. Ecora proposes the use of multi channel analysis of surface waves (MASW) over the proposed dam location to better understand to the profile of the known shallow bedrock. MASW has been previously used at Stocking Lake Dam however the locations do not provide adequate information for detailed design. MASW is described in more detail in section 2.5.2 below. Ecora will use manual hand augers to confirm MASW results and provide increased detail on the encountered soils overlaying the shallow bedrock. A standpipe piezometer will be installed in one of the hand auger boreholes to monitor fluctuations in water levels within the existing water supply pipe trench downstream of the existing dam.

2.5.2 Multichannel Analysis of Surface Waves (MASW)

MASW is a seismic method which utilizes low frequency surface waves to determine the shear wave velocity with respect to depth. Surface waves of different wavelengths travel in various velocities depending on the soil material. The wave velocities will reflect the changes in stiffness in the soil material. It is a non-intrusive in-situ test, very similar to other seismic data collection methods. A series of geophones are placed in a line along the surface of the target area. Multiple shots, or recordings, are taken to record the time offset and amplitude of the waves. Due to the close relation between surface wave velocity and shear wave velocity, the cross section between depth and shear wave velocity can be determined.

2.5.3 Existing Stocking Lake Dam Geotechnical Investigations

John Motherwell & Associates. – "Geophysical Survey Stocking Lake Dam" dated Jan 1975.

Seismic and resistivity geophysical survey data was performed by Geo-Recon Explorations Ltd. This indicates the bedrock profile along the existing dam structure.

EBA Engineering Consultants Ltd. – "Stocking Lake Dam Geotechnical Assessment" dated March, 2007.

On August 23, 2006, EBA arranged for a tracked mounted drill rig to advance four boreholes comprising augers and dynamic cone penetration testing to depths ranging between 2.1 and 10.4 mbgl. Standpipe monitoring wells were installed in two of the boreholes upon completion.

Ecora - "Dam Safety Review and Risk Assessment of Stocking Lake Dam" dated Nov 28, 2018.

This comprised of two boreholes with SPT testing, one on the upstream and one on the downstream side of the existing dam crest, using the sonic drilling method reaching depths of 5.2 m and 4.6 m respectively. Standpipe piezometers were installed in each borehole upon completion. To further supplement this investigation, Conetec Investigation Ltd. performed a geophysical survey comprising two two-dimensional

Multichannel Analysis of Surface Waves (MASW) tests to identify the location of the LLO and to determine the localized bedrock profile.

Ecora - "Geotechnical Assessment of Stocking Dam" dated June 2021.

The 2021 investigation comprised of a 22.5 mbgl inclined borehole using the sonic drilling method. This borehole was advanced adjacent to the outlet channel, downstream of the existing dam, at 45 degrees towards the spillway channel (azimuth of 245 degrees). This was completed to investigate the soil conditions under the spillway channel and to confirm the sharp dip in the bedrock profile as indicated in previous geophysical investigations. Hand augers and Lightweight Dynamic Cone Penetration (LDCP) tests were also performed to a maximum depth of 2.99 mbgl.

2.6 Environmental Assessment

Ecora's Qualified Environmental Professionals (QEP's) will confirm relevant details in the existing environmental assessments which investigates and catalogs the environmental assets at the dam site and in the surrounding areas and prepare a Construction Environmental Management Plan (CEMP) for the works.

- Ecora will work in-house with the environmental team at multiple stages to ensure the strict adherence to all necessary environmental regulations. Ecora will maintain communication with project stakeholders, including regulatory authorities, resource managers, and applicable community groups or local conservation organisations to ensure the project meets the requirements of all parties, and
- A guide to the preparation of a detailed site restoration and enhancement plan to be implemented post-construction to mitigate and offset any negative project impacts.

The environmental team will include a summary on any sensitive areas or protected ecosystems that must be considered during the detailed design development and construction phases.

The environmental team will provide support during construction phases of the project by providing environmental monitoring on behalf of the ToL, directing the implementation and effectiveness of mitigation measures, and documentation compliance with permit terms and conditions for the duration of the works. The environmental team will also manage any restoration and compensation requirements, to ensure all permit conditions are satisfied. Monitoring plans will be developed to ensure the environmental impact will be minimized.

2.7 Detailed Design

2.7.1 General

The emergency works performed in 2023 as summarized in Ecora's Stocking Lake Emergency Repairs Post-Construction Memorandum dated December 2023, were in response to seepage observed at the right abutment and softening of the dam embankment material. The emergency repairs were intended to address the aforementioned issues until such a time the dam could be replaced.

The detailed design will serve to elevate the preliminary design prepared by Ecora to a tender-ready design for construction and will include detailed cost estimates to assist in the final budgeting process. This phase will also include the preparation and submittal of any required provincial and federal permits for construction and an application for amendments to the Town's water license(s) as applicable.

2.7.2 Geotechnical Design

The geotechnical design scope of work generally encompasses several key aspects. One primary task is a detailed design stage review of the preliminary design considering the proposed 2024 investigation data to ensure accuracy and feasibility. This review includes, but is not limited to, several critical components:

- Seepage Analysis: Reviewing and refining the existing seepage analysis is essential to accurately assess and manage water flow through the dam structure. This involves evaluating current models, incorporating new data, and ensuring that the design effectively mitigates potential seepage issues.
- Slope Stability Assessments: Reviewing and refining existing slope stability assessments ensures that the dam slopes remain stable under various conditions, including static, dynamic, and seismic loads. This involves reassessing slope designs, performing additional stability analyses, and making necessary adjustments to enhance safety.
- Sheet Pile Wall Design: The existing sheet pile wall design must be reviewed and refined to ensure it adequately addresses seepage control and structural stability. This includes verifying penetration depths, material selection, and the wall's ability to withstand loads from the dam and water pressure.
- Construction Recommendations: This includes guidance on material placement, compaction requirements, construction sequencing, and measures to mitigate potential construction-related issues.

2.7.3 Structural Design

The scope of the structural design includes the detail design of the concrete portions of the structure including the spillway, concrete wing wall and concrete gate well structure. This will encompass the design of the steel reinforcement, the stability analysis of the concrete spillway structure, and the specification of the FRP sheet pile wall.

In addition to the concrete elements, the design will also include the steel bridge structures on the spillway and gatewell as well as bearing plates and connection of the bridges to the concrete structures. This also includes the specification of the guardrails and decking.

The structural design will include a Finite element analysis (FEA) using SAP2000 will be undertaken to evaluate the interaction between the FRP wall and the concrete structure. This work will be completed in collaboration with the geotechnical/hydrotechnical design team to determine applied design loads to ensure that the design meets/exceeds the minimum criteria as set forward in the following design guides and applicable criteria.

- 2007 CDA Dam Safety Guidelines and associated Technical Bulletins (2013 Edition).
- BC Dam Safety Regulation (BC. Reg. 40/2016).
- British Columbia Building Code 2024.
- National Building Code 2022.
- CSA/CAN A23.1 Design of Concrete Structures.
- CSA/CAN S16.1 Design of Steel Structures.

The analysis will also utilise and expand previous similar design cases such as the White River hydropower project completed in 2016 for the design of a FRP sheet piles with concrete footings.

The analysis can then be compared against the on-site instrumentation installed on the sheet piles to confirm that the assumptions made in the design model are accurate and the dam is behaving as expected on site.

As part of the role as structural engineers, an independent design review will be carried out by an alternative structural engineer as well as an independent review by an engineer not associated with the project to meet the EGBC criteria for structural quality control and assurance. Unless required otherwise, a type 1 as defined by EGBC Guide to the standard for documented independent review of high-risk professional activities or work, will be carried out on the design.

2.7.4 CADAM3D Concrete Structure Stability Analysis

A stability review of the detailed spillway structure design will be assessed in accordance with the CDA Dam Safety Guidelines (2007) that provides acceptance criteria for the structural stability of concrete gravity dams including the position of the resultant force for rotational modes of failure, the allowable normal compression stress, and minimum factors of safety for resistance to sliding for concrete gravity dams. This was already completed as part of the preliminary design stage, so will only require updating as per any detailed design changes. The stability review of the spillway will be undertaken utilizing the software CADAM3D v.2.5.11.1. The software performs the stability analysis of gravity dams, spillways, weirs, and water intakes based on the gravity method using rigid body equilibrium and beam theory to perform stress analyses, compute crack lengths and factors of safety for the static and seismic stability in accordance with the Canadian state of practice as documented in CDA (2007,2013) and USACE (1995).

2.7.5 Design Deliverables

Ecora will produce detailed design drawings for the proposed dam as refined through Ecora's findings during the previously completed preliminary design process. The detailed design drawings will include a level of detailed sufficient to construct the proposed works and may include but not limited to:

- Site maps displaying plan and cross-section views of key areas pre- and post-construction,
- Construction material specifications and volumes,
- Temporary dewatering or diversion works plan (if needed),
- Access areas, staging areas, and soil disposal areas (anticipated to be similar to the areas identified and utilized during the Stocking Lake Dam Emergency Repairs),
- Re-vegetation / environmental restoration plan(s),
- Constructability notes and details, and
- Reporting/Permitting/Licensing as required.

Ecora will complete a Plan Submission Report in accordance with Plan Submission Requirements for the Construction and Rehabilitation of Small Dams (Version 14a – July 2018). The Plan Submission Report serves as the primary permitting requirements to be submitted to the Dam Safety Officer (DSO) for the proposed works.

2.8 Tendering / Request for Proposals

Ecora understands the ToL anticipates construction to begin during the summer of 2025. Ecora will assist as requested by the ToL in the tendering phase of this project. Due to the complexity of the construction works, we are recommending that the ToL proceed with a Request for Proposal (RFP) process to procure construction services for this project, as opposed to traditional tender call. This will allow the ToL to evaluate the suitability of

potential contractors for this type of project and their proposed methodology for construction. The tasks for this phase will include but are not limited to:

- Producing a schedule of tender items and a Class A cost estimate,
- Assisting the ToL with the preparation and issuance of tender / RFP documents,
- Answer bidders' questions and prepare/issue addendums as necessary,
- Review bids for compliance and suitability to complete the work with the ToL solicitor if needed,
- Completing a decision matrix to assist the ToL with selection of a preferred construction contractor, and
- Issuance of contract documents for execution of the ToL and the Contractor.

2.9 Construction Support Services

The construction phase of the project is proposed to proceed as follows:

- Following the issuance and execution of the construction contract, Ecora will send a notice to proceed to the Contractor, initiating the construction phase,
- Ecora will organize a pre-construction meeting with the Contractor, Site Inspector, Contract Administrator, and the ToL in order to confirm the following;
    - Contact Information,
    - Permit Requirements,
    - Transportation to Site and any Vehicle/Foot Traffic Management Plans,
    - Communications,
    - Scope of Work,
    - Optimum Construction Staging,
    - Materials Testing, and
    - Roles and Procedures.
- Ecora will provide inspection services and contract administration services, as required, to ensure works are constructed in compliance with the contract documents and to meet compliance with IFC drawings,
- Inspection services shall include the following:
    - Keep complete construction records,
    - Inspect works, notify the contractor in writing of deviations from the specifications,
    - Monitor results of all material and quality control testing,
    - Issue all site instruction in writing within 24 hours from issuing such instruction. A copy of all instructions is to be kept for record purposes. The contractor will be required to coordinate and pay for all required quality control and materials testing (Ecora can provide quality control and testing upon request by the contractor).

- Contract Administration services shall include:
    - Geotechnical supervision, erosion and sediment control, and environmental (QEP) monitoring as required. Ecora expects that full-time environmental monitoring may be stipulated by one or more permits (see 3.11 Environmental Monitoring below),
    - Ecora will request random checks of line and grade constructed by the contractor and take measurements for the computation of construction quantities,
- Ecora's project manager will ensure all budgets, schedules and quality control specifications are met,
- Chair bi-weekly site meetings with the Contractor, Project Manager, Site Inspector, and the ToL Project Manager. The objective of the meetings shall be to maintain project communications, assess work progress, ensure compliance with construction schedule, identify potential change orders or extra work requirements, and, where possible, resolve disputes. This meeting shall also be used to document the number of working days utilized to date and any additions or deletions to the approved number of days to complete the Contract works.
- Preparation, certification, and submission of monthly progress payments at the end of each calendar month. A construction schedule, prepared by the contractor and acceptable to the Contract Administrator, is to be included in all progress payments; and,
- Issuance of substantial and total performance documentation at the conclusion of the construction tasks.

2.9.1 Strain Gauge Installation

The fibre reinforced sheet pile wall will require strain gauge installation to allow for measurement of load, stress and pressure. The layout, number and location for the installation of the strain gauges shall be confirmed following the detailed design and in consultation with the instrumentation supplier. This will be to confirm that the assumptions made during the design phase are accurate and the dam is behaving as predicted. Installation of the gauges will be performed by the suppliers technical personal, and this cost is not included in our fees. For preliminary purposes, it is anticipated that three locations will be used for the strain gauges with three gauges per location but may change following detailed design and consultation.

2.10 Environmental Monitoring

The Environmental team will provide support during the construction phase of the project by developing environmental management and mitigation plans for contractors, providing environmental oversight on behalf of the town, directing the implementation and effectiveness of mitigation measures, and documentation compliance with permit terms and conditions for the duration of the works. The environmental team will also manage the restoration and compensation requirements, to ensure all permit conditions are satisfied, which may include a maintenance period. This will be completed by Ecora's biologists and will include details on how the impacts to the environment will be monitored during construction. Ecora will oversee the implementation of the terms and conditions described in the environmental permits.

Ecora's environmental team will undertake an environmental impact evaluation of the required vegetation removal and identifying other environmental issues that will need to be addressed during the construction of the dam. This will begin during the detailed design phase but may continue into the construction phase.

Monitoring plans will be developed to ensure the environmental impact be minimized. A description of potential enhancement opportunities will be developed and provided.

The environmental monitoring and reporting work does not include any works associated with a Fisheries act authorization or offsetting required as part of an authorization that may arise from environmental impact reviews.

2.11 Post-Construction Services

The contractor will complete the post-construction survey. Survey data will be provided to Ecora for use in the record drawings.

Ecora will provide post-construction reports as required by the BC Dam Safety Regulation and will include a summary of the construction process and any deviations from the design. At the completion of the construction works, guidance and training will also be provided to the ToL staff to familiarize them with the operational and maintenance requirements of the dam.

A record drawings package will be provided to the ToL from Ecora.

3. Summary of Deliverables

The key deliverables for the detailed design phase include:

- Plan Submission Report that meets the requirements of the BC Dam Safety Regulations, which includes:
    - Development Report,
    - Design Drawing Package and Specifications (Issued for Permit),
    - Construction Supervision Plan and Schedule,
    - Material Specification Report,
    - Construction Environmental Monitoring Plan (CEMP),
    - Draft Operation, Maintenance and Surveillance (OMS) Manual, and
    - Draft Dam Emergency Plan (DEP).
- Class A cost estimate and schedule of quantities.

It is assumed that the design drawings, specifications, and cost estimate would be provided to the ToL for review and comment at 50% and 95% and the Plan Submission Report would be submitted as 100% detailed design draft for review prior to finalizing.

The key deliverables post construction include:

- Final commissioning report summarizing the construction records, recommendations, photo documentation, assessment of overall quality standards achieved etc.,
- Record drawings (will be included in the final commissioning report package), and
- Inspection of the work one month prior to the end of the warranty period and submit report to the ToL. It is understood that Ecora will also be responsible for any additional post-construction inspections required by permit(s) (e.g. to monitor habitat restoration).

All deliverables are to be signed and sealed in accordance with EGBC guidelines, following the ToL and Regulator review and acceptance of the documents (where applicable).

4. Ecora Personnel

4.1 Key Personnel

The Ecora team will comprise of senior engineer Michael J. Laws, P.Eng., and intermediate engineer Adam Kerk-Hecker, P.Eng. They will be supported by intermediate and junior engineers as required. The senior biologist, Adam Patterson, R.P.Bio., will be supported by intermediate and junior and biologists as required. Dr. Adrian Chantler, Ph.D, P.Eng., will support Ecora during this project as a senior/independent hydrotechnical reviewer. Dr. John Sully, Ph.D, C.Eng., P.Eng., will support Ecora during this project as a senior/independent geotechnical reviewer. This project will be led from our Kelowna office, with resources drawn from other offices, including Ecora's Vancouver office, as necessary.

The proposed Dam Safety Engineer has an excellent working relationship with the MoF Dam Safety Officers through his continued dam safety work experience across the province. A short description of the proposed team members and their abilities and duties follows, with detailed resumes of key personnel attached in Appendix A.

Michael J. Laws, P.Eng., Principal Dams & Geotechnics (Lead Qualified Professional) — Michael J. Laws has over 24 years of broad geotechnical, hydrological and dam engineering experience, including analysis and design of soil and rock slopes, rock fall assessments, geotechnical design of shallow and deep foundations, seismic and liquefaction analyses, specialized analysis of existing dams, structural stability of concrete gravity dams, dam break analysis, flood routing including inundation and flood mapping, design and rehabilitation of dams and the design of geostructures. He previously worked for New Zealand's second largest dam owner where he ran the dam safety program. He has worked on numerous run-of-river hydro projects, designed and undertaken engineering assessments on dams and completed over 20 comprehensive dam safety reviews as the lead qualified professional over the last 5 years including the Okanagan Lake Regulatory System dams for the Ministry of Forests.

Adam Kerk-Hecker, P.Eng., Civil/Hydrotechnical Engineer (Project Manager) — Mr. Kerk-Hecker is a hydrotechnical engineer with previous experience on dam engineering assessments and remediation design. Adam has been involved in over 15 dam remediation projects in the last 5 years. Adam will be managing this assignment and has experience working on projects on Vancouver Island and is familiar with the local geography. Adam will be responsible for managing deliverables and project milestones. Adam has over 8 years of consulting experience working as a civil/hydrotechnical engineer. His experience includes hydrology, hydraulics, numerical hydraulic modelling, and hydrological assessments of dam watersheds. Adam is well versed in both the BC Dam Safety Regulation and the Canadian Dam Association (CDA) Dam Safety Guidelines. Adam has experience using specialized hydrotechnical software such as HEC-RAS and HEC-HMS.

Adrian Chantler, Ph.D., P.Eng., Senior Hydrotechnical Engineer — Dr. Chantler is a hydrotechnical and civil Engineer with over 40 years of experience in the areas of hydrology, hydraulics, and physical and numerical hydraulic modelling. Adrian is a senior/independent reviewer. He has been engaged in floodplain studies and hydrotechnical engineering for dams and water resource projects throughout western and northern Canada. Dr. Chantler's experience encompasses river engineering, dam engineering, stormwater and watershed management, floodplain management, mine water management and dam safety reviews in North America and overseas. He has been involved with over 50 dam safety reviews in the last 10 years. He is an independent consulting engineer under contract to Ecora and will provide senior hydrotechnical review for this assignment.

Lalinda Weerasekara, Ph.D., M.A.Sc., P.Eng., Senior Geotechnical Engineer — Mr. Weerasekara is a senior geotechnical engineer who has previously worked on design for some of the largest civil projects in the Yukon,

including being the Geotechnical Engineer on record for the Crooked Creek and Nisutlin Bay Bridge projects. Lalinda's background experience includes seismic and stability analysis on embankments, deep and shallow foundations, and seismic liquefication analysis from high-consequence dams. He is proficient with multiple seismic modelling softwares, including D-MOD2000, SHAKE2000, DeepSoil, ProSHAKE, CLiq, SHAKE91, SeismoSignal and SeismoMatch.

Adam Patterson, R.P.Bio., Senior Biologist — Mr. Patterson is a Senior Biologist with 18 years of relevant consulting experience. His experience includes conducting and overseeing the implementation of environmental assessment and management for large infrastructure projects throughout BC. He has expertise in terrestrial and aquatic environments as they relate to fish and wildlife ecology, impact assessment, land development, and conservation. His experience includes various terrestrial wildlife and vegetation surveys and habitat assessments, species at risk assessments, habitat restoration and design, and environmental permitting, compliance monitoring, and auditing. Adam has provided environmental management support for dam safety reviews and inspections for Lind Creek, Holland Lake Dam, Stocking Lake Dam, Cannell Lake Dam, Dickson Lake Dam, and a series of dams for the Ministry of Forests near Okanagan Lake.

Duncan Hendricks, R.P.Bio., CPM, Senior Biologist — Duncan Hendricks is a Senior Biologist with 30 years of experience in environmental consulting throughout western and northern Canada. Duncan specializes in managing and directing multi-disciplinary projects. Duncan's technical background is in the assessment of aquatic habitat and aquatic restoration. He has extensive experience in assessing stream and river habitat throughout BC and the western United States.

Matt Roche, P.Eng., Structural Engineer – Mr. Roche is a Structural Engineer with 10 years of relevant consulting experience. Matt has previously worked on several hydro-geotechnical projects and dam assessments ranging from small structures such as the Chain Lake dam assessment and replacement to the very large such as the Hells gate arch wall dam assessment. He specialises in structural finite element modelling and dynamic analysis of structures. He is also the recipient of the 2009 Sean Du Courcy reward from the Concrete society of Ireland for his research into the durability of concrete with the addition of Carbon multiwalled nanotubes. He will be providing input and review on structural factors with context to the dams.

Patrick Machibroda, M.Sc., P.Eng., **Geotechnical Engineer —** Mr. Machibroda as over 7 years of experience in geotechnical engineering. He has worked on various large-scale projects in YK, BC, AB, SK, MB, ON, and NU. Patrick has worked with Ecora on the Chain Lake Dam Upgrades, Princeton, BC, the Conuma (Shikano) Tailings Storage Facility, Tumbler Ridge, BC, and the BC Hydro Feeder Extension. Patrick's areas of expertise include materials testing, drilling, slope stability analysis, deep foundation design, 2D Limit Equilibrium and Finite Element Modelling (slope stability, deformations/settlements, and seepage), and site inspections and reviews.

4.2 Supporting Personnel

Prajakta Jadhav, Ph.D., E.I.T., Geotechnical Consultant — Ms. Jadhav is a geotechnical consultant based in Ecora's Vancouver office. Her background is in civil engineering with a specialization in geotechnical engineering with experience in static and seismic design of geotechnical structures and numerical modeling of soil-structure interactions. She is proficient in a number of modeling and simulation software packages including: OpenQuake, OpenSees, ABAQUS, GEO-STUDIO, GEO-5, and MSEW. Ms. Jadhav has published multiple papers on seismic loading and will assist Ecora in the geotechnical data review.

Melony Catana, AScT, Environmental Technologist — Melony Catana is a registered Applied Science Technologist that has been consulting in BC for 18 years. She specializes in environmental assessment and mitigation for infrastructure and land development projects. She is experienced with conducting terrestrial and aquatic wildlife and vegetation surveys, habitat assessments, impact assessments, environmental management, project design, regulatory liaison, permitting and compliance monitoring. She has developed and implemented project-specific management plans for mitigating impacts to Species at Risk (SAR), including wildlife exclusion and salvages. She has extensive experience working as part multidisciplinary project teams for the remediation of

dams, culverts, and bridges, including fish and wildlife salvages and monitoring watercourse diversions. Her design experience includes fish and wildlife habitat restoration and compensation. She has worked extensively in permit acquisitions. Melony has provided environmental management support for dam remediations for Frazer Lake Dam, Yellow Lake Dam, Rose Lake dam and Star Gulch Dam.

Adam Tieman, E.I.T., Junior Hydrotechnical Engineer — Mr. Tieman is a Hydrotechnical Engineer-In-Training with 3 years of relevant consultant experience. Adam has experience using specialized hydrotechnical software such as FLO-2D with QGIS and HY-8. He is familiar with dam inundation mapping through his experience being involved in several ongoing safety reviews being completed for Mamit Lake, Becher, Toosey, and Botanie Lake dams.

Daniel Tamas, G.I.T., Junior Geoscientist — Mr. Tamas is a Geoscientist-in-Training (G.I.T.) with 6 years of experience with Ecora. Daniel's experience is split between the environmental and geotechnical sectors. Daniel has been coordinating geotechnical drilling programs and assessment reporting for major highway and pipeline projects across the interior. Daniel's area of expertise is in safety coordination, contractor communications, procurement of subcontractors and machinery support, field report development, terrain and geohazard mapping, and Standard Penetration Test (SPT) analysis and data processing. Mr. Tamas has experience communicating with First Nation stakeholders on a number of projects.

Abbas Rahman, PMP, Hydrotechnical Consultant — Mr. Rahman is an experienced Civil and Hydrotechnical engineer with Ecora. Abbas has over 11 years of experience. A chartered professional engineer in New Zealand, Abbas is currently a candidate for his P.Eng. with EGBC. Mr. Rahman is based out of Vancouver, BC, and has experience in hydrotechnical assessments. Abbas has worked on multiple culverts, runoff treatment, and bridge assessments. His work includes updating Botanie Lake, Toosey, and Becher dam OMS manuals and DEPs.

Bram Samuels, M.Eng., E.I.T., Junior Geotechnical & Hydrotechnical Engineer — Mr. Samuels has 4 years of industry experience in geotechnical and hydrotechnical engineering sectors. Bram is based out of Vancouver, BC and is experienced in modeling software including LiqSVs, HEC-RAS, HEC-HMS, FLO-2D, and MODFLOW. Bram has worked on projects including Seton Portage Debris Flow Mitigation, Nunns Creek Culvert Replacement, and Fairbanks Road Landslide Remediation. Bram is experienced with soil conditions on Vancouver Island, designing monitoring wells and piezometers.

Meghan Sherwood, E.I.T., Junior Geotechnical Engineer – Meghan Sherwood has experience conducting ARD/ML assessments and site reconnaissance, geotechnical site investigations, terrain analysis and mapping, slope stability analysis, retaining wall design, lab testing and construction monitoring. She is proficient with industry standard geotechnical slope stability and seepage modelling software such as GeoStudio Suite, Rocscience (Slide2, RocPlane, RocFall, SWedge, Dips) and gINT.

Tomos Edmonds, E.I.T., Junior Hydrotechnical Engineer — Mr. Edmonds 2 years of total experience in Ecora's civil, geotechnical and hydrotechnical departments. Tomos specializes in 2D/3D AutoCAD Civil3D drafting, has experience using specialized hydrotechnical software such as HEC-RAS, and supports the structural stability assessment of concrete gravity dams using CADAM3D. Tomos has worked on projects in the Ladysmith area on Vancouver Island including the Holland Creek Dam Safety Inspections & Mackie Rd Dam Decommissioning.

5. Achievements on Projects

5.1 Project Experience

Within the past ten years, Ecora has undertaken numerous dam projects throughout British Columbia. The scopes and scales of the projects vary and includes dam safety inspections, dam safety reviews, engineering assessments and designs, decommissioning plans, and dam construction and administering.

Ecora has experience working on a number of dams on Vancouver Island and in the Ladysmith area. These projects include the Holland Creek Dams Assessment and Decommissioning, Stocking Lake Dam Preliminary Design, Holland Lake Dam Safety Inspections, and Cowichan Valley Regional District DSRs for Stocking Lake Dam, Youbou Creek Dam, Ashburnham Creek Dam, and Shawnigan Lake Weir.

The Ecora team is experienced in undertaking dam safety reviews, engineering assessments of dams, conceptual, preliminary, and detailed dam design, and engineering services related to construction, with a successful reputation for completing similar types of assignments and acceptance of their reports by regulator(s). A list of dam projects that Ecora has completed is listed below. Full project profiles including detailed descriptions, client name and contact, scope, and scheduled/actual completion date(s) for selected recent dam projects are provided in Appendix D. See Table 5.1.1 below for a summary of projects Ecora has conducted since 2015.

Table 5.1.1 Past Dam Projects Completed by Ecora

Client	Province/ Territory	Dam	Services	Year
Ministry of Forests	BC	Nicklen Lake Dam	Comprehensive Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, geotechnical stability analysis, hydrotechnical assessments, reporting. Conceptual design of improvements to Nicklen Lake Dam.	2021-2023
City of Kelowna	BC	Frazer Lake Dam	Detailed design and construction services for the remediation of Frazer Lake Dam including design of a hybrid sheet pile embankment dam, LLO, and spillway, and procurement and contract administration services.	2019-2023
City of Prince George	BC	Shane Lake Dam	Detailed design, procurement, contract administration, and construction services for the remediation of Shane Lake Dam. Includes design of remediation works, including spillway upgrades, to address deficiencies identified during a Dam Safety Review.	2021-2022
ISL Engineering	BC	Centre Star Gulch Dam	Detailed design and construction services for the remediation of Centre Star Gulch Dam. Includes design of remediation works, including spillway upgrades, to address deficiencies identified during a Dam Safety Review.	2020-2022
Town of Ladysmith	BC	Stocking Lake Dam	Preliminary Design (60%) services for a replacement dam structure at Stocking Lake. Included the design of a hybrid sheet pile embankment dam, primary spillway structure with stilling basin, and low-level outlet works.	2021-2022
City of Abbotsford	BC	Cannell Lake Dam	Detailed design, procurement, contract administration, and construction services for the remediation of Cannell Lake Dam including re-design of a spillway, spillway control structure, and filter berm.	2020-2022
City of Powell River	BC	Haslam Lake Dam	Comprehensive Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, geotechnical stability analysis, hydrotechnical assessments, reporting.	2021
City of Penticton	BC	Ellis #4 Dam, Greyback Dam, Penticton #2 Dam	Comprehensive Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, geotechnical	2021

Client	Province/ Territory	Dam	Services	Year
			stability analysis, hydrotechnical assessments, reporting. Project was completed jointly with Austin Engineering Ltd.	
Ministry of Forests	BC	Yellow Lake Dam	Detailed design, procurement, contract administration, and construction services for the remediation of Yellow Lake Dam. Includes design of remediation works, including spillway upgrades, to address deficiencies identified during a Dam Safety Review.	2020-2021
Yukon Energy Corporation	YT	Aishihik Generating Station, Whitehorse Rapids Generating Station, Mayo Generating Station	Completion of audit style dam safety reviews on three generating stations owned and operated by Yukon Energy Corporation. DSR include review of eight structures of a variety of dam types including concrete, rockfill and earthen embankments.	2020-2021
City of Prince George	BC	Shane Lake Dam	Comprehensive Dam Safety Review including geotechnical stability and piping assessments of embankment dam. Consequence Classification Review including dam break analysis and inundation mapping. Preparation of Dam Emergency Plan and Operation, Maintenance & Surveillance Manual.	2020
City of Trail	BC	Violin Lake Dam	Consequence Classification Review including dam break analysis and inundation mapping. Recommended spillway improvements. Preparation of Dam Emergency Plan and Operation, Maintenance & Surveillance Manual.	2019-2020
Ministry of Forests	BC	Minton Lake Dam	Owner's Engineer during construction of new dam.	2017-2019
Ministry of Forests	BC	Okanagan Lake Dam, Skaha Lake Dam, McIntyre Dam	Comprehensive Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, structural stability analysis of concrete gravity dams, reporting. Includes condition assessment of vertical drop structures, drainage and dike works. Preparation of Dam Emergency Plan and Operation, Maintenance & Surveillance Manual.	2018-2019
City of Abbotsford	BC	Cannell Lake & Dickson Lake Dams	Engineering Assessment, incorporating dam geotechnical investigation, geotechnical stability assessment of embankment dam, reporting. Includes predesign of rehabilitation works.	2018-2019
Ministry of Forests	BC	Swan Lake Dam	Dam operations plan that provides guidance on outflow.	2019
Cowichan Valley Regional District	BC	Stocking Lake Dam, Youbou Creek Dam, Ashburnham Creek Dam, Shawnigan Lake Weir	Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, structural stability analysis of concrete gravity dams, geotechnical stability and piping assessments of embankment dam, reporting. Preparation of Dam Emergency Plan and Operation, Maintenance & Surveillance Manual.	2018-2019
City of Trail	BC	Cambridge Reservoir Dam	Consequence Classification Review including dam break analysis and inundation mapping. Preparation of Dam Emergency Plan and Operation, Maintenance & Surveillance Manual.	2018-2019

Client	Province/ Territory	Dam	Services	Year
Regional District of Okanagan-Similkameen	BC	Chain Lake Dam	Consequence Classification Review of Chain Lake Dam including dam break analysis and inundation mapping.	2018
No. 21 Great Projects Ltd. (The Ponds)	BC	Hill Spring Dam	Completion of plan submission report for the rehabilitation of Hill Spring Dam. Includes design of new embankment and spillway weir. Preparation of Dam Emergency Plan and Operation, Maintenance & Surveillance Manual.	2016-2018
Xaxli'p First Nation / Neskonlith Indian Band / Cook's Ferry Indian Band / Indigenous Services Canada	BC	Kwotlenemo Lake Dam / Neskonlith Lake Dam / Calling Lake Dam	Dam Safety Reviews incorporating dam break analysis and inundation mapping, consequences classification review, geotechnical stability and piping assessments of embankment dams, reporting. Preparation of Dam Emergency Plan and Operation, Maintenance & Surveillance Manual.	2016-2017
Ministry of Forests	BC	Yellow Lake Dam	Engineering Assessment, incorporating dam break analysis and inundation study, geotechnical investigation, geotechnical stability assessment of embankment dam, reporting.	2016-2017
Regional District of North Okanagan	BC	Goose Lake Dam	Engineering Assessment, incorporating dam break analysis and inundation study, geotechnical investigation, geotechnical stability assessment of embankment dam, reporting.	2016
The City of Abbotsford	BC	Cannell Lake & Dickson Lake Dams	Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, geotechnical stability and piping assessments of embankment dams, reporting.	2015-2016
Xat'sull First Nation / Indigenous Services Canada	BC	Rose Lake Dam	Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, geotechnical stability and piping assessments of embankment dam, reporting.	2015-2016
Ministry of Forests	BC	Swan Lake Dam	Engineering Assessment, incorporating dam break analysis and inundation study, geotechnical investigation, geotechnical and structural stability assessment of gravity dam, reporting.	2015-2016
Allendale Water Users Community	BC	Allendale Lake and Clark Meadows Lake Dams	Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, geotechnical stability and piping assessments of embankment dams, reporting.	2015
City of Rossland	BC	Centre Star Gulch Dams	Dam Safety Review incorporating dam break analysis and inundation mapping, consequence classification review, geotechnical stability and piping assessments of embankment dams, reporting.	2015

5.2 Client References

Table 5.2.1 below provides two references and corresponding projects that Ecora has completed where the Michael J. Laws, P.Eng. was the Lead Qualified Professional (LQP).

Table 5.2.1 List of References for Recent Dam Projects Where Michael J. Laws, P.Eng. was the LQP

Project	Project Date /Duration	Tasks	Client	Client Contact	Client Contact Position	Contact
Shane Lake Dam Remediation	2020-2021	▪ Detailed Design ▪ Plan Submission Report ▪ Construction Administration ▪ Procurement ▪ Construction Services	City of Prince George	Alan Clark, P.Eng.	Infrastructure Engineer	250-614-7826 Alan.Clark@prince george .ca
Centre Star Gulch Dam Remediation	2021-2022	▪ Detailed Design ▪ Plan Submission Report ▪ Construction Services	ISL Engineering and Land Services Ltd.	Sean Annan, P.Eng.	Project Engineer	250-362-2229 Sannan@islengine ering.com

6. Project & Quality Management

6.1 Project Management

For this assignment Ecora's Project Manager will be Mr. Adam Kerk-Hecker, P.Eng. who will work closely with the Dam Safety Engineer to ensure all engineering aspects of the design and construction are carried out effectively and efficiently. He will also be responsible for undertaking many of the hydrotechnical aspects of the project. Mr. Kerk-Hecker has worked on previous projects on Vancouver Island including the conceptual and preliminary design, and emergency remediation of Stocking Lake dam, and Holland Creek Dam Safety Inspections and Decommissioning. Mr. Kerk-Hecker is a professional engineer with strong communication skills and has managed many similar projects. It is anticipated that the project manager shall:

- Liaise with the Town, licensees, and the BC MoF DSO to access available information and verify DSO requirements for review, inspection, and reporting,
- Provide brief monthly updates, more frequent meetings during phases on the critical path, to the Town on work status, schedule, and budget. Meetings are expected to be virtual unless an in-person meeting is required,
- Identify potential risks to scope, schedule, and budget as early as possible and take appropriate steps to mitigate such risks by working collaboratively with the Town,
- Coordinate with all internal and external disciplines and sub-contractors/consultants to ensure project efficiency, and
- Plan arrangements for meetings, recording of meeting minutes, and tracking of action items.

The meetings will be conducted whenever possible remotely to mitigate costs. Meeting minutes shall be circulated to all participants for comment and confirmation and copied to the respective project officials as soon as possible.

6.2 Quality Management

Ecora's quality control processes are contained within our Professional Practice Management Plan (PPMP). This is a living document that is updated as required for policy changes and is signed off by all Responsible Registrants.

All projects begin with a Project Start-up meeting that includes all team members. Roles and Responsibilities of each team member are clearly defined and any barriers to project delivery are discussed and addressed. Project Objectives, Deliverables and Critical Success Factors are documented, and a review schedule is determined.

All team members are required to sign off on the completed Project Start-up form and this form is filed on the network in the Project Folder. If new team members are brought on to the team, they are required to review the project start-up form and sign off their understanding. During the scheduled review as defined in the project start-up form, success milestones are assessed, and any deficiencies are discussed amongst team members to drive performance improvements.

Additionally, small, office-based discipline team meetings are held on a weekly basis. During these meetings, lessons learned from on-going or completed projects are discussed. While topics may be broad and include lessons learned on professional development, client relations, technical subjects, and safety, the discussion of all aspects of project delivery helps to ensure performance improvement in the long-term.

6.3 Document Control

Ecora's record and document management are split into three categories; setting up project filing, preparing documents, and filing documents. These three categories are detailed below:

Setting up Project Filing

A standard project directory is created for each project and tagged with the internal project number.

Preparing Documents

Standard templates are used to facilitate consistent quality, and Ecora uses validated and approved software & media for creating and maintaining documents. Standard file naming conventions to save document files are used and include document identifiers (project name, project number, filename, file directory) in the document, as appropriate, or in document properties or metadata stored with each electronic file.

The project name, project number, purchase order number, and topic are included in the subject line of project or work-related e-mail containing information. Peer review, spellcheck and check documents are applied before submittal.

Filing Documents

All documents and email communications are filed in their appropriate directory, with correct revisions in the name of the document. Periodically, files are converted and saved to a portfolio PDF and saved to the project filing. Printed documents are converted to digital documents and saved to files. A check-in/check-out system is in place where multiple users have access to working documents. Included is a revision record indicating revision number, what was revised and by whom on documents where version control is required (drawings, reports, etc.). Revisions are clearly identified and documented with version control.

Issuing Documents

An issue record is included, indicating purpose for issuing and when issued, on documents where version control is required (drawings, reports, etc.). The revision and issue records may be combined into one record. When issuing electronic documents, providing and retaining the file in PDF/A exactly as issued. A formal transmittal form or informal transmittal (form, e-mail or other) may be used as a record of what was sent to whom, when and how.

Receiving Documents

All documents received from the client are stored in the project file, designated as 'received documents'. If receiving physical documents, the document gets coded and filed in the digital document project file structure. On an as-needed basis, actions taken will be recorded based on received documents.

7. Safety

Ecora is committed to compliance with all government agencies, regulators and regulations, and industry best practices. Our Health and Safety Program follows a safety-first philosophy for best practices utilizing companywide policies, rules, procedures, and continued on-going training. Our safety program is externally audited and certified by the BC Forest Safety Council. We provide our employees with ongoing training, and information on company policies, safe work, and emergency procedures.

Prior to undertaking fieldwork, Ecora staff are required to complete a Job Hazard Assessment (JHA) and an Emergency Response Plan (ERP) for review and approval by the Ecora project manager. The JHA summarizes any potential and known hazards associated with the work site and type of work to be undertaken, the personal protective requirements, training requirements, and work procedures and mitigation methods to be followed by Ecora staff. Additionally, the JHA lists the required external personnel such as flaggers and utility locators. The ERP further highlights the risks associated with field work and must be signed by all Ecora employees who visit the site.

Ecora is Safety Accord Forestry Certified (SAFE).

Ecora's WorkSafeBC Registration number is 954444 and is currently in good standing with WorkSafeBC.

8. Schedule and Fees

It is envisioned that the geotechnical site investigations can be performed concurrently with the site investigations for the Holland Lake project, with a target window of summer/fall 2024. This could reduce the overall cost for this phase for the detailed design of Stocking Lake. For the construction support services fees, the period of construction was assumed to be 12 weeks.

The detailed design phase of the project would be performed during the winter of 2024/2025. The target submission date for the Plan Submission Report is spring 2025.

It is estimated the fees and disbursements for Ecora to undertake this project is estimated at $381,439, excluding GST. See Appendix C for a fee and disbursement breakdown.

Table 8.1 Fee Summary (excluding GST)

Phase	Sub-Total
Project Management	$11,040
Site Investigation and Background	$21,716
Detailed Design Development / Permitting	$149,830
Pre-Construction	$20,755
Construction*	$158,050
Post-Construction	$20,048
Total:	**$381,439**

*Note: Engineering support fees for the construction phase are based on an estimated 12-week construction timeline. Actual fees for this phase are dependent on construction and are billed at time and materials.

9. Closure

We trust this proposal meets your present requirements. Do not hesitate to call with any questions. We look forward to the opportunity of working with you on this project.

Sincerely,

Ecora Engineering & Environmental Ltd.

Prepared by:



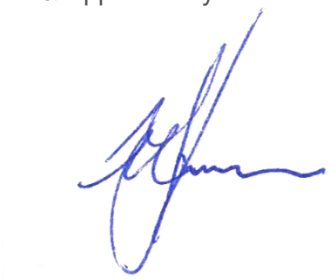
Tomos Edmonds, E.I.T.
Junior Hydrotechnical Engineer
Direct Line: 431.761.1894
tomos.edmonds@ecora.ca

Reviewed by:



Adam Kerk-Hecker, P.Eng.
Hydrotechnical Engineer
Direct Line: 250.469.9757 x1073
adam.kerkhecker@ecora.ca

Reviewed & Approved by:



Michael J. Laws, P.Eng.
Principle Dams & Geotechnics Engineer
Direct Line: 250.469.9757 x1045
michael.laws@ecora.ca

References

Ecora. (2022). *Stocking Lake Dam Remediation Preliminary Design*.

Appendix A

Resumes

CONSULTANT PROFILE

Michael J. Laws, P.Eng.

Senior Geotechnical & Dam Safety Engineer

Summary of Experience

Mr. Laws is a Senior Geotechnical and Dam Safety Engineer with over 24 years' experience in dam engineering, including undertaking dam safety reviews of existing dams, geotechnical investigations, design and construction supervision of new dams and specialized analysis of existing dams and appurtenant structures. He has successfully completed comprehensive dam safety reviews, design and engineering assessments of over 30 dams in the past 5 years and is very familiar with the requirements of the Canadian Dam Association, Dam Safety Guidelines 2007 (2013 edition) and the 2016 BC Dam Safety Regulation. He is experienced with specialized geotechnical, structural stability and hydrotechnical modelling software for modelling embankment stability, seismic performance, seepage, settlement, dam breach and inundation such as Geostudio Suite, Slide, Plaxis 2D & 3D, CADAM, L-Pile, Shake, BREACH, and FLO—2D.

Areas of Experience

- Embankment Dam Engineering and Dam Safety Reviews
- Structural Stability of Concrete Gravity Dams
- Deep and Shallow Foundation Design
- Geotechnical Site Investigation and Design
- Numerical Modelling with Specialization in Finite Element Analysis
- Retaining Wall Design
- Seismic Hazard Assessments and Analysis
- Stability of Soil/Rock Slopes and Landslide/Rock Fall Remediation
- Landslide Runout and Impulse Wave Analysis
- Dam Breach Analysis
- 2D Hydraulic Modelling for Flood Plains and Open Channels

Relevant Experience

- Comox Dam Safety Review, Courtenay, BC (2023—Ongoing) – Project Manager and Lead Qualified Professional Engineer for an audit-style Dam Safety Review of Comox Dam in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and the BC Dam Safety Regulation. DSR report prepared for BC Hydro.
- Lind Creek Dam Safety Inspection, Greenwood, BC (2023—2024) – Lead Qualified Professional Engineer for a Dam Safety Inspection involving site reconnaissance and preliminary hydrological analyses.
- Mamit Lake Dam Safety Review, Lower Nicola, BC (2022—2024) – Lead Qualified Professional Engineer for the comprehensive dam safety review of Mamit Lake Dam in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and the BC Dam Safety Regulation. The scope of work included undertaking dam inspections, dam breach and 2D flood routing analyses, consequences classification review, structural stability analysis, preparation of Dam Emergency Plans and Operation, Maintenance &



Contact Information

michael.laws@ecora.ca

Education

Bachelor of Engineering (Civil), University of Auckland, New Zealand

Bachelor of Science (Geology), University of Auckland, New Zealand

Years of Experience

24+

Years with Firm

9+

Office

Kelowna, BC

Associations

Canadian Dam Association

Canadian Geotechnical Society

Member Association of Professional Engineers and Geoscientists of Alberta (APEGA)

Member Association of Professional Engineers and Geoscientists of British Columbia (APEGBC)

Member Association of Professional Engineers and Geoscientists of Saskatchewan (APEGS)

Member Engineers Yukon

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Surveillance manuals, and preparation of DSR report. A Dam Safety presentation was provided to dam operators at the conclusion of the project.

- Haslam Lake Dam Safety Review, Powell River, BC (2021) – Lead Qualified Professional Engineer for the comprehensive dam safety review of Haslam Lake Dam in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and the 2016 BC Dam Safety Regulation. A Dam Safety presentation was provided to dam operators at the conclusion of the project.
- Violin Lake Dam Safety Review, Trail, BC (2019) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety review of Violin Lake Dam in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and the 2016 BC Dam Safety Regulation. The scope of work, included undertaking dam inspections, dam breach and 2D flood routing analyses, consequences classification review, structural stability analysis, preparation of Dam Emergency Plans and Operation, Maintenance & Surveillance manuals, decommissioning cost estimate, design of spillway improvements, and preparation of DSR reports.
- Minton Lake Dam Rehabilitation, Williams Lake, BC (2018—2019) – Owners Engineer for the rehabilitation of Minton Lake Dam, including review of existing back ground information, development of the technical scope of work for the RFP, assistance in evaluation of the bids, and review of technical deliverables for the Ministry of Forests, Lands, Natural Resource Operations and Rural Development.
- Dam Safety Reviews of the Okanagan Lake, Skaha Lake and McIntyre Dams, Okanagan Valley, BC (2018—2019) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety reviews of the of the Okanagan Lake, Skaha Lake and McIntyre concrete gravity dams in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and the BC Dam Safety Regulation. The scope of work, included undertaking dam inspections, dam breach and 2D flood routing analyses, consequences classification review, structural stability analysis, preparation of Dam Emergency Plans and Operation, Maintenance & Surveillance manuals, and preparation of DSR reports for the BC Ministry of Forests, Lands, Natural Resource Operations and Rural Development.
- Cannell & Dickson Lake Dams Engineering Assessment, Mission, BC (2018—2019) – Lead Engineer for the engineering assessment of two existing embankment dams, including undertaking an intrusive geotechnical investigations and vibrating wire piezometer installation, updated liquefaction triggering analysis, piping assessment, seepage and stability analysis, conceptual design of remedial works and reporting for the City of Abbotsford.
- Chain Lake Dam Consequences Classification Re-Determination, Princeton, BC (2018) - Lead Qualified Professional Engineer for the consequence's

Training

APEGBC Technical Seminar on Methods for Predicting Metal Leaching and Acid Rock Drainage (ARD) Potential , Vancouver, Canada - December 2015

PLAXIS Advanced Course on Computational Geotechnics & Dynamics – Berkeley, USA – May 2013

PLAXIS Advanced Course on Computational Geotechnics – New York, USA – May 2012

PLAXIS Short Course on Computational Geotechnics & Dynamics – Portland, USA – August 2011

Professor Harry Polous Piling Short Course, Canterbury University, Christchurch, NZ – September 2006.

Geo-Slope International Geotechnical Modelling Workshop (including Slope/W, Seep/W and Sigma/W) Monash University, Melbourne, Australia – July 2003

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classification re-determination of an existing embankment dam, including undertaking dam inspection, dam breach and 2D flood routing analyses, consequences classification review, and reporting for the Regional District of Okanagan-Similkameen.

- Dam Safety Reviews and Risk Assessment of the Stocking Lake, Ashburnham Creek, Youbou and Shawinigan Lake Dams, Vancouver Island, BC (2018—2019) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety reviews of the of the Stocking Lake, Ashburnham Creek, Youbou and Shawinigan Lake Dams in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and the BC Dam Safety Regulation. Stocking Lake Dam is an earthen embankment, while, Ashburnham Creek, Youbou and Shawinigan Lake Dams are concrete gravity dams. As part of the scope of the DSR an intrusive geotechnical investigation including installation of piezometers of Stocking Lake Dam was undertaken. The scope of work, included undertaking dam inspections, dam breach and 2D flood routing analyses, consequences classification review, static and dynamic geotechnical stability analyses, structural stability analysis, preparation of Dam Emergency Plans and Operation, Maintenance & Surveillance manuals, risk assessments in accordance with the National Disaster Mitigation Program framework and preparation of DSR reports for the Cowichan Valley Regional District.
- Cambridge Dam Consequences Classification Re-Determination, Trail, BC (2018) – Lead Qualified Professional Engineer for the consequences' classification re-determination of an existing embankment dam, including undertaking dam inspection, dam breach and 2D flood routing analyses, consequences classification review, and reporting for the City of Trail.
- Hill Spring Dam Rehabilitation, Kelowna, BC (2017—2018) – Lead Design Engineer for the rehabilitation of an existing embankment dam, including undertaking intrusive geotechnical investigations, dam breach and 2D flood routing analyses, consequences classification review, seepage and stability analyses, design of embankment rehabilitation options, dam preparation of design reports, drawings and specifications for the Ponds.
- Dam Safety Reviews of Calling Lake, Neskonlith Lake and K'wotlenemo Lake Dams, Southern Interior, BC (2017) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety reviews of the three earthen embankment dams in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and BC Dam Safety Regulation, including dam inspections, dam breach and 2D flood routing analyses, consequences classification review, static and dynamic stability analyses, piping assessment, preparation of Dam Emergency Plans and Operation, Maintenance & Surveillance manuals and preparation of DSR reports for Indigenous and Northern Affairs Canada.

Michael J. Laws, P.Eng.
Senior Geotechnical & Dam Safety Engineer

- Allendale Lake Dam Geotechnical Assessment, Okanagan Falls, BC (2018) – Lead Engineer for the geotechnical assessment of an existing embankment dam, including undertaking an intrusive geotechnical investigations and piezometer installation, liquefaction triggering analysis, piping assessment, seepage and stability analysis, for the Allendale Waters Users Community.
- Yellow Lake Dam Engineering Assessment, Penticton, BC (2016—2017) – Lead Engineer for the engineering assessment of an existing embankment dam, including undertaking an intrusive geotechnical investigations and piezometer installation, dam breach and 2D flood routing analysis, consequences classification review, seepage and stability analysis, piping assessment and reporting for the Regional District of North Okanagan.
- Goose Lake Dam Engineering Assessment, Vernon, BC (2016—2017) – Lead Engineer for the engineering assessment of an existing embankment dam, including undertaking an intrusive geotechnical investigations and piezometer installation, dam breach and 2D flood routing analysis, consequences classification review, seepage and stability analysis, pipping assessment and reporting for the Regional District of North Okanagan.
- Dam Safety Review of Cannell Lake and Dickson Lake Dams, Mission, BC (2015—2016) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety review of two embankment dams in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and BC Dam Safety Regulation, including dam inspection, dam breach and 2D flood routing analysis, consequences classification review, static and dynamic stability analysis, pipping assessment, landslide generated impulse wave analysis, preparation of DSR reporting for the City of Abbotsford.
- Dam Safety Review of Rose Lake Dam, Williams Lake, BC (2015—2016) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety review of the Rose Lake dam in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and BC Dam Safety Regulation, including dam inspection, dam breach and 2D flood routing analysis, consequences classification review, static and dynamic stability analysis, preparation of DSR reporting for Indigenous and Northern Affairs Canada.
- Dam Safety Reviews of Calling Lake, Neskonlith Lake and K'wotlenemo Lake Dams, Southern Interior, BC (2017) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety reviews of the three earthen embankment dams in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and BC Dam Safety Regulation, including dam inspections, dam breach and 2D flood routing analyses, consequences classification review, static and dynamic stability analyses, piping assessment, preparation of Dam Emergency Plans and Operation,

Michael J. Laws, P.Eng.
Senior Geotechnical & Dam Safety Engineer

Maintenance & Surveillance manuals and preparation of DSR reports for Indigenous and Northern Affairs Canada.

- Allendale Lake Dam Geotechnical Assessment, Okanagan Falls, BC (2018) – Lead Engineer for the geotechnical assessment of an existing embankment dam, including undertaking an intrusive geotechnical investigations and piezometer installation, liquefaction triggering analysis, piping assessment, seepage and stability analysis, for the Allendale Waters Users Community.
- Yellow Lake Dam Engineering Assessment, Penticton, BC (2016—2017) – Lead Engineer for the engineering assessment of an existing embankment dam, including undertaking an intrusive geotechnical investigations and piezometer installation, dam breach and 2D flood routing analysis, consequences classification review, seepage and stability analysis, piping assessment and reporting for the Regional District of North Okanagan.
- Goose Lake Dam Engineering Assessment, Vernon, BC (2016—2017) – Lead Engineer for the engineering assessment of an existing embankment dam, including undertaking an intrusive geotechnical investigations and piezometer installation, dam breach and 2D flood routing analysis, consequences classification review, seepage and stability analysis, pipping assessment and reporting for the Regional District of North Okanagan.
- Dam Safety Review of Cannell Lake and Dickson Lake Dams, Mission, BC (2015—2016) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety review of two embankment dams in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and BC Dam Safety Regulation, including dam inspection, dam breach and 2D flood routing analysis, consequences classification review, static and dynamic stability analysis, pipping assessment, landslide generated impulse wave analysis, preparation of DSR reporting for the City of Abbotsford.
- Dam Safety Review of Rose Lake Dam, Williams Lake, BC (2015—2016) – Project Manager and Lead Qualified Professional Engineer for the comprehensive dam safety review of the Rose Lake dam in accordance with the Canadian Dam Association Dam Safety Guidelines 2007 (2013 Edition) and BC Dam Safety Regulation, including dam inspection, dam breach and 2D flood routing analysis, consequences classification review, static and dynamic stability analysis, preparation of DSR reporting for Indigenous and Northern Affairs Canada.

CONSULTANT PROFILE

Adam Kerk-Hecker, P.Eng.
Civil/Hydrotechnical Engineer

Summary of Experience

Adam Kerk-Hecker is a Hydrotechnical Engineer based out of Ecora's Kelowna office. He has over nine years of experience in civil and hydrotechnical engineering with a Bachelor of Engineering in Civil Engineering from BCIT.

His experience has been primarily focused on small to medium embankment dam systems design, hydraulic structure design, watershed hydrological assessments, 1D/2D hydraulic modelling, dam safety reviews, culvert and open channel drainage design, streambank/lakeshore erosion protection design, and minor road design. Adam is proficient in hydrologic modelling and streamflow prediction in gauged or ungauged watersheds including using the latest projected data and industry standards to account for potential climate change impacts. He is familiar with various hydrologic/hydraulic software tools (1D/2D HEC-RAS, HEC-HMS, HY-8), project management, topographical and construction survey, contract administration, cost estimating, and reporting. He also has extensive experience with AutoCAD Civil3D modelling and design. Adam is well versed with the BC Dam Safety Regulations (BC Reg 40/2016) and 2007 Canadian Dam Association (CDA) Dam Safety Guidelines (2013 Edition).

Areas of Expertise / Specializations

- Hydrotechnical engineering design
- Dam engineering
- Hydrological analysis
- 1D/2D Hydraulic Modelling (HEC-RAS, HEC-HMS, HY-8)
- 2D/3D CAD drafting and modelling
- Contract administration
- Project management
- Cost estimating

Relevant Experience

Kitley Creek Dam Remediation – Detailed Design and Construction, Kaleden, BC (2021 – Ongoing)

- Conceptual/detailed design and construction services for improvements to Kitley Creek Dam in Kaleden, BC. Duties included hydrological analysis and modelling, flood routing, design of a new low-head concrete dam with integrated spillway and low-level outlet, production of design drawings/specifications, and provincial permitting requirements (Plan Submission Report). Hydrotechnical engineer of record.

Chain Lake Dam Upgrades – Dam Safety Review, Detailed Design, and Construction, Princeton, BC (2021 - Ongoing)

- Conceptual/detailed design and construction services for improvements to Chain Lake Dam near Princeton, BC. Duties included hydraulic modeling of



Contact Information
adam.kerkhecker@ecora.ca

Office
Kelowna, BC

Education
B.Eng. in Civil Engineering from the British Columbia Institute of Technology – 2017

Diploma in Civil Engineering Technology from the British Columbia Institute of Technology – 2014

Associations / Memberships
Engineers and Geoscientists of British Columbia (EGBC)

Years of Experience
+9

Years with Firm
+7

Adam Kerk-Hecker, P.Eng.
Civil/Hydrotechnical Engineer

the IDF, flood/reservoir routing, design of a new ogee spillway control structure with stilling basin, design of a replacement low-level outlet, upgrades to the spillway channel and dam embankment, production of design drawings/specifications, and provincial permitting requirements (Plan Submission Report). Hydrotechnical engineer of record.

Frazer Lake Dam Rehabilitation – Detailed Design and Construction, Kelowna, BC (2019 – 2022)

- Design and construction services for the reconstruction of Frazer Lake Dam in Kelowna, BC. Duties include hydrological analysis and modelling of the watershed and IDF, design of a hybrid embankment dam with a sheet pile centre wall, design of a low-level outlet intake structure and pipe, production of design drawings/specifications, Plan Submission Report, procurement services, field reviews, and post-construction reporting.

Shane Lake Dam Improvements – Dam Safety Review, Detailed Design, and Construction, Prince George, BC (2021 – 2022)

- Design and construction services for improvements to Shane Lake Dam in Prince George, BC. Duties included hydrological analysis and modelling, flood routing, design of a new open channel lined spillway, production of design drawings/specifications, Plan Submission Report, procurement services, contract administration, field reviews, and post-construction reporting.

Stocking Lake Dam Replacement – Dam Safety Review, Conceptual Design, and Preliminary Design, Ladysmith, BC (2020 – 2022)

- Dam Safety Review, Conceptual Design, and Preliminary Design for the replacement of Stocking Lake Dam near Ladysmith, BC. The existing earthen embankment dam at Stocking Lake was determined to contain multiple deficiencies and needs to be replaced. The preliminary design consists of a new FRP sheet pile/rockfill dam downstream of the existing dam, RC spillway structure and stilling basin, and RC gated intake tower. Duties include hydraulic modelling of the dam, hydrologic analysis and modelling of the watershed, preliminary design of the spillway control structure, spillway channel, intake tower, dam geometry, and access road. Design included the production of preliminary design drawings/specifications, design report, and Class C cost estimate.

Yellow Lake Dam Upgrades – Dam Safety Review, Detailed Design, and Construction, Kaleden, BC (2020 – 2022)

- Dam Safety Review, design, and construction services for upgrades to the Yellow Lake Dam in Kaleden, BC. Duties include hydraulic modelling of the dam, spillway, and downstream channel, design of a spillway control structure, design of spillway channel upgrades, production of detailed design drawings/specifications, Plan Submission Report, procurement services, contract administration, and post-construction reporting.

Cannell Lake Dam Rehabilitation – Detailed Design, and Construction, Abbotsford, BC (2019 – 2020)

- Completed detailed design, reporting, and permit acquisition for the rehabilitation of Cannell Lake Dam in Abbotsford, BC. Duties included review of hydrological analysis of the watershed and IDF, hydraulic (1D/2D) modelling of IDF passage through dam, hydraulic design of the spillway control structure and channel, hydraulic design of the dam structure, surface modelling in CAD, production of detailed design drawings, Class A construction cost estimates, production of Plan Submission Report, tender support, contract administration, and post-construction services.

Dickson Lake Dam and Cannell Lake Dam Rehabilitation Conceptual Design, Abbotsford, BC (2018 – 2019)

- Completed conceptual design and reporting for the rehabilitation of Dickson Lake Dam and Cannell Lake Dam in Abbotsford, BC. Duties included review of hydrological analysis of the watershed and IDF, hydraulic (1D/2D) modelling of IDF passage through dam, hydraulic design of the spillway control structure and channel, preliminary hydraulic design of the dam structure, surface modelling in CAD, production of design drawings, construction cost estimates, and production of preliminary design report.

CONSULTANT PROFILE

Adrian Chantler, Ph.D., P.Eng.

Senior Hydrotechnical Engineer

Summary of Experience

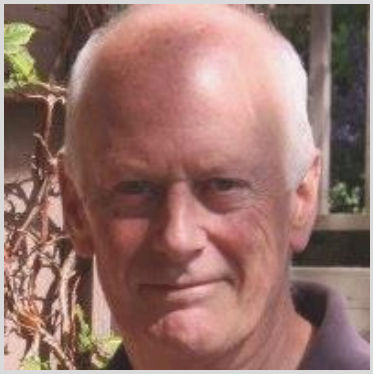
Dr. Chantler is a Civil Engineer with over 40 years of experience in the areas of hydrology, hydraulics, and physical and numerical hydraulic modelling. He has been engaged in floodplain studies and hydrotechnical engineering for dams and water resource projects throughout western and northern Canada. Dr. Chantler's experience encompasses river engineering, dam engineering, stormwater and watershed management, floodplain management, mine water management and dam safety reviews in North America and overseas. He has been involved with over 60 dam safety reviews in the last 10 years. He is an independent consulting engineer under contract to Ecora.

Areas of Experience

- Dam safety reviews
- Floodplain mapping
- Bridge and culvert hydraulic design
- Mine water management
- Protection of pipeline river crossings
- Hydrological studies
- River engineering
- Physical model studies of hydraulic structures
- Research on hydraulic geometry of tidal watercourses

Relevant Experience

- Hydrotechnical and senior review for the Haslam Lake Dam safety review (City of Powell River, 2021-Ongoing)
- Hydrotechnical and senior review for the Haslam Lake Dam safety review (City of Powell River, 2021)
- Senior review for dam safety reviews for seven Yukon Energy Corporation dams (Yukon Energy Corporation, 2020-2021)
- Hydrotechnical and senior review for the Shane Lake Dam safety review (City of Prince George, 2020)
- Hydrotechnical review for dam safety reviews and risk assessments for Stocking Lake, Ashburnham and Youbou Dams and Shawnigan Lake Weir (Cowichan Valley Regional District, 2018-2019)
- Hydrotechnical review for dam safety reviews for three Okanagan Lake Regulation System dams (Ministry of Forests, Lands, Natural Resource Operations and Rural Development, 2018-2019)
- Development of mitigation options for dam breach scenarios at Sooke, Saddle and Deception Dams (Capital Regional District, 2018)
- Hydrotechnical component of Clayton Falls Dam safety review (BC Hydro, 2018)
- Senior review of Niskonlith Lake Dam safety review (Neskonlith Indian Band, 2017)



Contact

adrian.chantler@ecora.ca

Education

Ph.D., Civil Engineering Hydraulics, University of Strathclyde, Scotland, 1973

B.Sc. (Hons.), Civil Engineering, University of Dundee, Scotland, 1970

Years of Experience

40 +

Associations

Canadian Dam Association

Member Association of Professional Engineers and Geoscientists of British Columbia (APEGBC)

Member, Canadian Water Resources Association (CWRA) (BC President 1996/7)

Work History

Independent Consulting Hydrotechnical Engineer, under contract to several consulting firms 2014 to present

Tetra Tech EBA/EBA Engineering Consultants Ltd.
Manager of the Water and Marine Engineering Group, then Principal Specialist 2004–2014

Adrian Chantler, Ph.D., P.Eng.
Senior Hydrotechnical Engineer

- Senior review of dam breach inundation study for Tabor Lake Weir (Regional District of Fraser-Fort George, 2017)
- Senior review of Kwotlenemo Lake Dam safety review (Xaxli'p Indian Band, 2017)
- Senior review of Calling Lake Dam safety review (Cook's Ferry Indian Band, 2017)
- Senior review of Yellow Lake Dam engineering assessment (MFLNRO, 2016)
- Hydrotechnical component of Sooke Lake Dam safety review (Capital Regional District, 2016)
- Senior review of Goose Lake dam engineering assessment (Regional District of North Okanagan, 2015)
- Senior review of Dickson and Cannell Lake dam safety reviews (Abbotsford Mission Water Commission, 2015)
- Senior Review of Rose Lake dam safety review (Xaťsūll First Nation, 2015)
- Senior review of Swan Lake dam safety review (North Okanagan Regional District, 2015)
- Senior review of Allendale and Clark Meadows dam safety reviews (Blue Mountain Vinyards, 2015)
- Senior review of Arrowsmith Dam inundation study (Arrowsmith Water Service, 2014)
- Senior reviewer for Cumberland Creek dam's inundation study (Village of Cumberland, 2014)
- Member of technical committee on removal of Fortymile Creek Dam (Town of Banff, 2014)
- Senior reviewer for hydrotechnical aspects of Brent and Farleigh Dam Safety Reviews (Aboriginal Affairs and Northern Development, 2014)
- Senior reviewer for hydrotechnical aspects of Nickel Plate Tailings Facility Dam Safety Review (Barrick Gold Inc., 2014)
- Senior reviewer for hydrotechnical aspects of Saddle Lake Dam Safety Review (Regional District of Kootenay Boundary, 2013)
- Senior reviewer for hydrotechnical aspects of HB Tailings Storage Facility Dam Safety Review and remedial design of spillway chute and energy dissipator (RD of Central Kootenay, 2013/14)
- Review of dam breach analyses and inundation maps for Postill, Bulman, South and McKinley Dams, (Glenmore Ellison Improvement District, 2012/13)
- Senior reviewer for hydrotechnical aspects of Lacey Lake No.1 Dam Safety Review (Cherry Creek Water District, 2012)
- Hydrotechnical aspects of dam safety audits for Battle River Dam and Sheerness Cooling Pond Dyke, (ATCO Power, AB, 2012)
- Dam Safety Review project management for Lund and Thulin Lake Dams, (Lund Waterworks Improvement District, 2012)

Hay & Company Consultants Ltd., Principal, later President, of a specialist water resources consulting company that was acquired by EBA 1997–2004

Steffen Robertson and Kirsten, Principal in Hydrotechnical Engineering, 1988–1997

Crippen Consultants, Hydraulic Design Engineer then Head of the Hydrotechnical Department 1975–1988

Crouch and Hogg in Glasgow, Scotland, Assistant Engineer 1973–1975

CONSULTANT PROFILE

LALINDA WEERASEKARA, PH.D., P.ENG.
SENIOR GEOTECHNICAL ENGINEER

Summary of Experience

Lalinda Weerasekara is a Senior Geotechnical Engineer with over 18 years' experience in geotechnical engineering in both research and consulting throughout Western Canada. He has extensive experience working on projects in the Yukon Territory for the past 7 years and was the Geotechnical Engineer on Record for several of them.

Lalinda has completed both a Masters and a Doctorate in Civil Engineering, and his experience with evaluating seismic hazards is evidenced by his leading the section on seismic design of foundations for the 2023 Canadian Foundation Engineering Manual. He is familiar with a wide range of modelling software's used for evaluating embankment and slope stability, seismic performance, settlement including D-MOD2000, SHAKE2000, DeepSoil, ProSHAKE, CLiq, SHAKE91, SeismoSignal, SeismoMatch (Seismic), SLIDE, GEOSTUDIO suite (slope stability) and SETTLE3D (settlement). Lalinda is also well versed in dam related geotechnical work including evaluating liquefaction and slope/embankment stability and has completed designs for projects on the Alaska Highway where permafrost was a high concern geohazard.

Areas of Experience / Specializations

- Seismic Ground Response and Liquefaction Assessments
- Seismic Ground Deformation Analyses
- Shallow and Deep Foundations Design
- Slope Stability and Retaining Walls Design
- Ground Improvements Design and Recommendations
- Utility Analysis for Seismic/landslide and other Loadings
- Settlement and Preload Design
- Geotechnical Site Characterization and Advanced Laboratory Testing
- Cut-and-Cover Tunnel

Relevant Experience

- Walter Hardman Dam, Revelstoke, BC, Canada (2019 - ongoing) -Undertake seismic liquefaction triggering and seismic ground deformation analyses as a part of the Dam Safety Review. As a part of this analysis, 2D FLAC model was developed to estimate the static bias coefficient. Client: BC Hydro model.
- Louise Lake Dyke, Whitehorse, YK, Canada (2016) - Development of input ground motions for the seismic assessment, site-specific ground response analyses and liquefaction assessment, 2D FLAC (seismic ground deformation) analysis for the high consequence dam. Client: ATCO.
- Clinton Creek Waste and Tailing dumps, YK, Canada (2016): * Preliminary seismic liquefaction analyses. Client: Government of Yukon.
- Farleigh Lake Dam, Penticton, BC, Canada (2016) - Site-specific seismic liquefaction analyses, developing ground motions for seismic analyses and review of slope stability analyses. Client: Penticton Indian Band.



Contact Information

lalinda.weerasekara@ecora.ca

Education

Ph.D. Civil Engineering, University of British Columbia - 2011

M.A.Sc., Civil Engineering, University of British Columbia - 2007

B.A.Sc. Civil Engineering, University of Peradeniya, Sri Lanka - 2002

Years of Experience

18+

Years with Firm

<1

Office

Vancouver, BC

Associations

Member Association of Professional Engineers and Geoscientists of British Columbia (APEGBC)

Member of Engineers Yukon

LALINDA WEERASEKARA, PH.D., P.ENG.
SENIOR GEOTECHNICAL ENGINEER

- Bralorne Mine, BC, Canada (2016) -Seismic liquefaction analyses, review of stability analyses, filter design and remediation options to improve the stability. Client: Bralorne Gold Mine Ltd.
- Cantung Mine, Tungsten, NWT, Canada (2012, 2015): * Conducted seismic liquefaction, seismic ground deformation and stability analyses for two of the existing and one proposed tailing storage facilities. Client: North American Tungsten Corporation Ltd.
- Bluefish Dam, Yellowknife, NWT, Canada (2012): * Review of grouting details and QA/QC work. Client: Northwest Territories Power Corporation (NTPC).
- Nisutlin Bay Bridge, Teslin, YK, Canada (2020 - ongoing): Geotechnical Engineer of Record. The project includes design and construction of replacing the existing 600 m long bridge over Teslin Lake. Lalinda is responsible for the seismic liquefaction and ground deformation assessments, pile design, slope stability, permafrost evaluations and providing reviews during construction. Client: Yukon Government.
- Nordenskiold River Bridge, Carmacks, Yukon, Canada (2019-2022). Geotechnical Engineer for the new bridge over the Nordenskiold River, which is part of the new Carmacks bypass road. The bridge was designed as a lifeline bridge. Responsible for the geotechnical analyses which included soil liquefaction, slope stability, frost jacking forces, axial and lateral pile capacity. A thorough review of the CAPWAP analysis was required for the 40 m long piles at the abutments due to complexities matching signals. Client: Yukon Government.
- Nares River Bridge, Carcross, YK, Canada (2016-2018): Geotechnical Engineer of Record. The existing bridge was located along Klondike highway near Carcross. The replacement bridge is a three-span structure of about 145 m long. Responsible for the developed earthquake ground motions, conduct seismic liquefaction, and ground deformation analyses. Performed lateral and axial pile capacity calculations, including the generation of p-y curves for structural analysis. Estimated settlements and conducted slope stability assessments for the approach fills. Client: Yukon Government.
- Crooked Creek Bridge Replacement and Highway Realignment, Stewart Crossing, YK, Canada (2018 –ongoing): Geotechnical Engineer of Record. The existing bridge located along Klondike highway south of Stewart Crossing. The bridge was selected replacement to improve the geometric design and increase speed to 100 km/h. Responsible for the settlement estimations, slope stability, seismic analyses, pile design and preparing specifications. The design challenges included soil liquefaction, permafrost, frost heaving, settlement, large cut volumes and high fill embankments. Client: Yukon Government.
- MacDonald Creek Bridge, Racing River Bridge and Tetsa River Bridge No 1, , Alaska Highway, BC (2021-ongoing). Review engineer for the conceptual design of the replacement bridges at these locations. The study included

evaluating the potential soil conditions/geology, seismic concerns, permafrost potential, stability of approach embankments and constructability concerns. Client: Client: Public Service Procurement Canada

- Bougie Creek slope stability, Alaska Highway, BC, Canada (2020): Review Engineer for the preliminary geotechnical assessment completed for the slope instability observed near Bougie Creek along the Alaska Highway. The area is potentially impacted by multiple geohazards, including permafrost, compressible soft soils, river erosion and prehistoric landslide debris. Preliminary recommendations for mitigating the slope instability were provided, which included construction of a piled wall or road realignment. Client: Public Services and Procurement Canada (PSPC).

Journal Publications

- Weerasekara, L. (2023). “Impact from bridge superstructure on pile foundations in lateral spreading areas”, International Journal of Bridge Engineering.
- Weerasekara, L, Hall, B.E., and Wijewickreme, D. (2017), “A new method for designing and analyzing rein-forced soil structures”, Geosynthetics International.
- Wijewickreme, D and Weerasekara, L. (2015) “Analytical Modelling of Field Axial Pullout Tests Performed on Buried Extensible Pipes”, Journal of Geomechanics, ASCE.
- Weerasekara, L. and Wijewickreme, D (2010) “An Analytical Method to Predict the Pullout Response of Geo-textiles”. Geosynthetics International. 17(4): 193-206
- Weerasekara, L. and Wijewickreme, D. (2008) “Mobilization of Soil Loads on Polyethylene Natural Gas Pipe-lines Subject to Relative Axial Ground Displacements”, Canadian Geotechnical Journal, 45(9):1237-1249

Conference Publications

- Weerasekara, L., Wang, T, Jiang, J., Zhang, Q. and Hamersley, B. (2023). “Seismic Performance Evaluation of Previously Retrofitted Oak Street Bridge”, Canadian Conference-Pacific Conference on Earthquake Engineering 2023, Vancouver, BC, from June 25-30, 2023.
- Weerasekara, L., Miller, C., Honegger, D. and Wijewickreme, D. (2023). “Semi-empirical framework to predict watermain failure rates from permanent ground displacement for a regional scale seismic vulnerability study”, Canadian Conference-Pacific Conference on Earthquake Engineering 2023, Vancouver, BC, from June 25-30, 2023.
- Weerasekara, L., Rahman, M. and Wijewickreme, D. (2023). “Performance of buried natural gas distribution pipes in a landslide area: Observations from nine years of pipe strain monitoring”, Canadian Geotechnical Conference, Saskatoon, Saskatchewan.

LALINDA WEERASEKARA, PH.D., P.ENG.
SENIOR GEOTECHNICAL ENGINEER

- Weerasekara, L. (2023). “Limitations in existing methods for analyzing MSE walls reinforced with steel strips, Geotechnical Symposium, Vancouver Geotechnical Society, Vancouver, BC.
- Weerasekara, L. (2022). “A simplified approach to model deck resistance and back rotation of abutments due to lateral spreading”, GeoCongress, ASCE, Charlotte, NC.
- Weerasekara, L. (2021). “Limitations in using full scale instrumented wall results to develop and validate de-sign methods for MSE wall”, Canadian Geotechnical Conference, Niagara, Ontario.
- Sinha, T., Dhar, A.S., Weerasekara, L., and Rahman, M. (2021). “Effects of lateral ground movements on MDPE gas distribution pipes and branches”, Canadian Geotechnical Conference, Niagara, Ontario.
- Weerasekara, L. and Rahman, M. (2019). “Framework for assessing integrity of natural gas distribution pipes in landslide areas”, Canadian Geotechnical Conference, St. Johns, Newfoundland.
- Reza, A, Dhar, A.S., Rahman, M. and Weerasekara, L (2019). “Pulling rate effects on the pullout force of buried small diameter MDPE pipe in loose sand”, Canadian Geotechnical Conference, St. Johns, Newfound-land.
- Zhang, Q, Weerasekara, L., Jianping, J. and Shahria, A.M. (2019) “Seismic performance of integral abutment bridges in liquefiable soils”, 12th Canadian Conference on Earthquake Engineering – CCEE 2019, June 2019, Quebec
- Weerasekara, L, (2018), “Improvements to Pullout Failure Estimation in MSE Walls”, Canadian Geotechnical Conference, Edmonton, AB.
- Weerasekara, L, (2018), “Steel Strip Reinforced Soil Walls at Working Stress Conditions”, Canadian Geotechnical Conference, Edmonton, AB.
- Weerasekara, L, Hall, B.E., and Wijewickreme, D. (2016), “Ultimate Load Carrying Capacity of Steel Strip Reinforced Earth Structures”, Canadian Geotechnical Conference, Vancouver, BC.
- El-Dean, G.D., Weerasekara, L., Azizian, A., Mills, R. and Oljace, G. (2016), “Equivalent-Linear and Nonlinear Effective Stress Analyses for Watermain Seismic Design”, Canadian Geotechnical Conference, Vancouver, BC.
- Hall, B.E., Weerasekara, L, and Dabeet, A. (2016), “Case History of Ground Heave and Lateral Movement Resulting from Pile Driving”, Canadian Geotechnical Conference, Vancouver, BC.
- Weerasekara, L., Hall, B.E., and Wijewickreme, D. (2015), “A New Approach to Assess Soil-Geosynthetic In-teraction in Reinforced Slopes and Walls”, 23rd Symposium of the Vancouver Geotechnical Society on Soil-Structure Interaction, Vancouver, BC.

LALINDA WEERASEKARA, PH.D., P.ENG.
SENIOR GEOTECHNICAL ENGINEER

- Azizian, A, Weerasekara, L., Hall, B.E., Naesgaard, E., Amini, A., Khan, S., Knaus, M., and Preece, M. (2015), “Seismic Soil-Structure Interaction Analyses for an Elevated Guideway Structure of the Evergreen Line”, 23rd Symposium of the Vancouver Geotechnical Society on Soil-Structure Interaction, Vancouver, BC.
- Rahman, M. and Weerasekara, L. (2014) “Field Monitoring and Modelling of Natural Gas Pipelines Subject to Ground Movement”, Geohazards6, Kinston, Ontario., June 15-18, 2014.
- Dinovitzer, A., Fredj, A., Rahman, M., Weerasekara, L. and Wijewickreme, D. (2014). “Performance Monitoring of Small Diameter MDPE Natural Gas Pipelines Subject to Ground Movement”, Proc. of 14th International Pipeline Conf., ASCE, Calgary, AB. September 29-October 3, 2014.
- Wijewickreme, D and Weerasekara, L. (2011) “Analytical Modelling of Pipe Subject to Axial Ground Displacement”, Proceedings of the 12th IACMAG Conference, May 9 - 11, 2011, Melbourne, Australia.
- Weerasekara, L. and Wijewickreme, D (2010) “Response of Buried Plastic Pipelines Subject to Lateral Ground Movement”, Proc. of 8th International Pipeline Conf., ASCE, Calgary, AB.
- Weerasekara, L. and Wijewickreme, D. (2010) “Full-scale Model Testing of Buried Extensible Pipes Subject to Relative Axial Soil Loading”, 7th International Conference on Physical Modelling in Geotechnics (ICPMG), June 28 - July 01, 2010, Zurich, Switzerland.
- Wijewickreme, D. and Weerasekara, L., (2009) “Pipeline Geotechnical Engineering, Encyclopedia of Lifeline Support Systems, www.eolss.net (online).
- Wijewickreme, D, Weerasekara, L and Johnson, G. (2008) “Soil Load Mobilization in Axially Loaded Buried Polyethylene Pipes”, Proceedings of the 12th IACMAG Conference, 1-6th Oct, Goa, India.
- Weerasekara, L., Wijewickreme, D. and Mitchell, A. (2006) “Response of Tee-Junctions in Buried Polyeth-ylene Natural Gas Distribution Piping Subject to Ground Movement”. In Proceedings of the 59th Canadian Geotechnical Conference, 1-4 October 2006, Vancouver, B.C.

CONSULTANT PROFILE

Adam Patterson, R.P.Bio

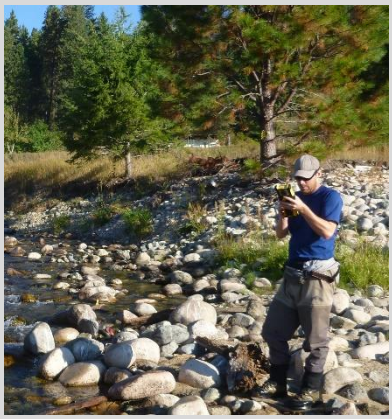
Environmental Impact Assessment Team Lead – Senior Biologist

Summary of Experience

Adam Patterson is a Registered Professional Biologist who has been conducting wildlife and fisheries fieldwork since 2000 and working in environmental consulting since 2005. His interests are focused on terrestrial and aquatic environments as they relate to fish and wildlife ecology, impact assessment, land development, and conservation. His experience includes various terrestrial wildlife and vegetation surveys and habitat assessments, species at risk assessments, ecosystem mapping and classification, and environmental permitting for a wide variety of small to large land development projects throughout BC. Adam manages many multi-disciplinary projects at Ecora, generally including environmental services such as biophysical inventory, impact assessment, and mitigation planning. Adam works closely with First Nations partners on many projects, including training and coordination of field crews, collaboration and sharing of Traditional and Cultural information, and attendance at engagement meetings and workshops.

Relevant Experience

- **BC Hydro Bridge River Transmission Project (BRTP) Environmental Services** – Adam has been the Principal Biologist and project manager of the environmental scope of services for the BC Hydro BRTP since early 2021. The project includes development of a Valued Components (VC) Table and workplan to conduct surveys for identified environmental values such as wildlife and species at risk, important habitat features and ecosystems, stream crossings and fish presence. Adam is part of the Szumin'ts project team, which is an Indigenous business and partnership between Ecora and St'at'imc First Nation communities. Adam's role includes coordination and planning with the archaeology and heritage team, as well as the overseeing the environmental services.
- **Fortis Interior Gas Upgrade (IGU)** – Adam is the project manager and technical lead providing environmental services and regulatory support for various projects related to the Fortis IGU program. To date these include drilling investigations, restoration programs, and contributing to project planning and mitigation strategies to avoid or reduce risks to aquatic and other sensitive habitats.
- **Westbank First Nation (WFN) Foreshore Inventory Update and Analysis** – Adam was the project manager and technical lead for the WFN foreshore mapping and update project. The project was initiated in 2022 and included a workplan to review existing Okanagan Lake FIM and WFN shoreline information, collect current and additional foreshore, shoreline, and riparian biophysical data, and provide spatial data results, including statistical analyses Spatial data tools were developed to facilitate use and interpretation of the data during WFN review of development applications, referrals, and other shoreline and foreshore planning efforts.
- **Tsal'alh Indigenous Guardians Program** – Adam is the lead biologist and project manager in support of the Tsal'alh Indigenous Guardians program, in the Lillooet region of BC. Adam led the program proposal and helped secure federal funding to support the Tsal'alh mule deer population study initiative,



Contact Information

adam.patterson@ecora.ca

Education

Bachelor of Science in Wildlife and Fisheries (2005); University of Northern British Columbia

Applied Science Diploma in Recreation Fish, and Wildlife Technology (1998); Selkirk College

Associations

College of Applied Biology:

Registered Professional Biologist

Training and Certificates

- Project Management (PSMJ)
- Occupational First Aid Level 1
- Electrofishing Crew Lead
- Riparian Areas Protection Regulation Assessment Course
- Power Systems Safety Program (PSSP)
- BC Hydro Heritage Awareness & Helicopter Safety Awareness
- Small Vessel Operator Proficiency (SVOP) & Marine Emergency Duty (MEDA3)
- Amphibian and Reptile Salvage Methods
- Ungulate Field Survey Methods
- Experimental Design in Ecology

Adam Patterson, R.P.Bio

Environmental Impact Assessment Team Lead – Senior Biologist

which includes development of a workplan and methodology to inventory and assess deer populations within the Tsal'alh Traditional Territory using Traditional Knowledge and Elder interviews. This project is being delivered through the Szumin'ts Indigenous Business and includes capacity building for the Tsal'alh Guardians, including GIS licensing and training, developing fieldwork data collection methods, and other technical support.

- **Conuma Willow Creek Expansion (WCE) Baseline Wildlife Inventory** – Adam is the project manager and technical wildlife lead for the WCE project in northern BC. The project includes determination of Valued Components in collaboration with First Nations, development of Wildlife Habitat Ratings (WHR) tables and wildlife inventory, as well as the coordination and completion of fieldwork. Adam will also lead the reporting, mapping, and data analysis for the project, expected to continue through 2023.
- **BC Hydro Site C Fish Stranding Monitoring Program (Mon-12)** – Adam was the project manager and technical lead for the BC Hydro Fish Stranding Monitoring Program (MON-12) along the Peace River near Fort St. John. The project was conducted over a four-year period, which included complex field and safety planning and coordination, conducting fish stranding monitoring surveys, data management and analysis, summary reporting, and mapping. The program is intended to determine the change in magnitude of fish stranding associated with the dam construction and focussed on baseline data collection.
- **Conuma Hermann Mine Site Reptile and Amphibian Salvage** – Adam was the project manager, senior biologist, and field coordinator for the amphibian and reptile salvage works for the Hermann Mine Test Sample project in northern BC. The project included a review of site conditions and footprint to determine risks to wildlife. Ecora developed a salvage plan and acquired Wildlife Act permits for the collection, handling, and translocation of identified species of concern, including frogs, toads, and snakes. Initial fieldwork and inventory was completed in fall of 2020 with additional monitoring efforts to be completed in spring 2021.
- **City of Kelowna Frazer Lake Environmental Assessment –** Adam is the project manager and technical lead for the environmental impact assessment and mitigation planning for proposed dam remediation works at Frazer Lake in Kelowna. Adam planned and completed biodiversity inventory surveys, including plant and ecosystem inventory, songbird point counts, owl call-playback and amphibian aural surveys, bat detection, and shoreline surveys. The inventory results were used to develop mitigation strategies which became part of an Environmental Management Plan for the project, which required review of potential dam safety improvement scenarios. Adam also oversaw the environmental permitting associated with the proposed works.

Workshops and Conferences

- College of Applied Biologists Conference (2023)
- Wetlands Institute – Wetland Restoration Techniques (2020)
- Qualified Environmental Professional Workshop - Species and Ecosystems at Risk (2020)
- Designing Avoidance and Mitigation for Migratory Birds and Species at Risk Critical Habitat Workshop (2019)
- Critical Habitat Screening Workshop (2019)
- Foreshore Plant Species At Risk Workshop (2018)
- Professional Practice Guidelines: Legislated Riparian Area Assessments in BC (2017)
- Riparian Areas Regulation Workshop, West Kelowna (2015)
- Wetlands, not wastelands Panel Discussion and Wetland Tour, Kelowna (2015)
- Wetland Keepers Workshop, Kelowna (2015)
- Wetland Working Group Workshop, Grand Forks (2014)
- Map Our Marshes Workshop, Peachland (2013)
- WildResearch Nighthawk surveys, Penticton and Kelowna (2012-2014)

Years of Experience

>20

Years with Firm

7

Office

Kelowna, BC

CONSULTANT PROFILE

Duncan Hendricks, R.P.Bio, CPM

Director of Environment – Senior Biologist

Summary of Experience

Duncan Hendricks is a Senior Biologist with 30 years of experience in environmental consulting throughout western and northern Canada. Duncan specializes in managing and directing multi-disciplinary projects. Duncan's technical background is in the assessment of aquatic habitat and aquatic restoration. He has extensive experience in assessing stream and river habitat throughout BC and the western United States.

In his previous role, Duncan supported BC Hydro's transmission and distribution projects throughout the province. In this role, he managed a dedicated team of environmental coordinators and specialists to deliver environmental, archaeological, and geotechnical deliverables.

Mr. Hendricks has experience in identifying and quantifying environmental impacts and outlining mitigation and compensation strategies associated with municipal infrastructure, transportation, mining, and hydroelectric developments. His experience includes management, coordination, and preparation of environmental assessments.

Relevant Experience

- **BC Hydro, Environmental Support for Transmission and Distribution Projects** – Program Manager for all environmental, geotechnical, and archaeological work as a subcontractor on Wood Canada's Engineering Services contract with BC Hydo. The projects included environmental and ground engineering work connected to upgrades and construction of new substations as well as transmission and distribution lines. The environmental support included archaeology, contaminants, aquatic, wildlife, and vegetation biology, and hazardous materials.
- **BC Hydro Victoria to Esquimalt Cable Replacement** – Overall supervision of environmental scope for 230 kV oil-filled cable replacement between Victoria and Esquimalt. The project included work within the urban area of Victoria, where the cable runs in ductbanks, as well as work within the marine environment of the Gorge area. Risk analyses were completed for shallow burial and directional drilling through the Gorge area. The work included environmental overview assessment, eelgrass survey, environmental management plan, and support for permitting and approvals.
- **BC Hydro 2L13/14 Upgrades** – Overall supervision of environmental scope for pole replacement on 230 kV transmission line in Capilano River watershed in North Vancouver. The project involved construction of new roads and work pads within a highly sensitive area, as well as helicopter access work. Work included environmental overview assessment, Marbled Murrelet surveys, management of archaeological and geotechnical services, environmental management plans, environmental monitoring and auditing, and *Water Sustainability Act* Section 11 permitting.
- **BC Hydro Kamloops Substation** – Overall supervision of environmental scope for construction of the new Kamloops Substation and associated line upgrades. Work included environmental overview assessment, management of

Contact Information

duncan.hendricks@ecora.ca

Education

Bachelor of Science in Biology and Physical Geography (1992); Carleton University

Certificate in Fish and Wildlife (1993); Lethbridge College

Associations

College of Applied Biology:

Registered Professional Biologist

Training and Certificates

- Certified Project Manager (PMLG)
- Occupational First Aid Level 1
- Certified Electrofishing Crew Lead, BC and USFW
- Riparian Areas Protection Regulation Assessment Course
- BC Hydro Power Systems Safety Program (PSSP)
- BC Hydro Heritage Awareness & Helicopter Safety Awareness

Workshops and Conferences

- College of Applied Biologists Conference (2024)
- Canadian Wildlife Service Migratory Bird Regulation Training (2023)
- River Restoration Northwest (2017)

Years of Experience

>30

Years with Firm

1

Office

Kamloops, BC

Duncan Hendricks, R.P.Bio, CPM

Director of Environment – Senior Biologist

archaeological and geotechnical services, environmental management plans, a detailed Lewis's Woodpecker management plan, and environmental auditing.

- **FortisBC Interior Transmission System Transmission Integrity Project** – Overall supervision of the environmental scope for FortisBC's ITS Transmission Integrity Management Capability (TIMC) Project. FortisBC identified the need for alterations to three sections of pipeline and 12 facilities. The project involved a review of biophysical and land use resources, an assessment of the potential effects on biophysical and/or land use resources, prescription of mitigation measures that could be used for required environmental protection, and development of a Class 3 cost estimate for environmental and archaeological costs associated with the implementation of the project.
- **TransMountain, TMEP Highway 1 Re-alignment Project** – Overall supervision of the environmental scope for Trans Mountain re-alignment of the eastbound lanes of Trans-Canada Highway No. 1 at two locations between Hope and Chilliwack, BC. The project included the following tasks conducting full-time environmental auditing to ensure compliance with the project construction environmental management plan (CEMP) and conformance with the environmental plans, instructions, and regulations, conducting regular communication, development of an approach for dealing with contact water from trenches within areas with potential for ML/ARD, and response to environmental incidents including spills to ensure compliance with the CEMP and regulations.
- **BC Hydro and Power Authority, Kitimat LNG Transmission Line Environmental Assessment** – Project manager, senior biologist, and reviewer for the environmental assessment for the construction of the transmission line between Terrace and Kitimat BC to support LNG Canada's facility in Kitimat. This scope involved an assessment of aquatic, wildlife, vegetation, and socio-economic resources, as well as a separate archaeological overview assessment. The project involved virtual open house meetings with First Nations to address concerns with the project.
- **Exxon, West Coast Canada LNG Environmental Assessment, Prince Rupert, BC** – Senior advisor for the Environmental Assessment through BCEAA and CEAA for Exxon's proposed LNG export facility. Proposed facility would include production, storage, and loading of LNG as well as transportation by vessel through BC waters to offshore markets. Role included management and support of staff and resources as well as interaction with client to ensure efficient project delivery.
- **City of Kelowna, Road Upgrade and Bridge Replacement Environmental Assessment, Mission Creek, Kelowna, BC** – Senior biologist for a detailed environmental impact assessment for the City of Kelowna road improvement project at Swamp, Casorso, and Benvoulin Roads. The road improvement project included replacing the existing bridge over Mission Creek, and road upgrades adjacent to several other streams, wetlands, and ditches. Conducted detailed fish and wildlife inventories of the areas to be impacted by the project. Prepared a habitat compensation with the City of Kelowna and the Ministry of Environment.

CONSULTANT PROFILE

Matthew Roche, P.Eng.
Structural Engineer

Summary of Experience

Matthew Roche is a structural engineer with the Ecora Team. Based in Kelowna, he is part of the Structural Team and supports the group with project management, the design of commercial and multi-story buildings, and dam engineering services. Mr. Roche supports clients in both the public and private sectors.

Matthew has worked on various high-profile projects worldwide, including as the Lead Design and Project Engineer on the Microsoft European Headquarters in Dublin, Ireland, and the SIT Gymnasium in Southland, New Zealand. He has also worked on multi-million-dollar housing projects and large data storage facilities throughout Europe. Mr. Roche has assisted Ecora with the structural analyses of multiple dam projects including Chain Lake Dam, Kitley Creek Dam, and Frazer Lake Dam.

His expertise includes FEA modelling, lightweight serviceability analysis of buildings and bridges, structural assessment of buildings, design of small and large-scale commercial, residential, and industrial buildings, and engineering analyses related to dam infrastructure.

He has experience with design of various materials including reinforced concrete, structural steel both mild and stainless, timber, masonry both unreinforced and reinforced, and composite materials, such as composite steel and concrete.

Matthew has also lectured in Advanced Engineering Design for the Master of Engineering program at the Dublin Institute of Technology.

Areas of Experience

- Large scale and small scale Commercial and Industrial buildings.
- Tilt up concrete building design.
- Steel, Concrete and Timber building design to Canadian and European design code standards.
- Seismic design and analysis of buildings in Canada and New Zealand.
- Structural design and analysis of concrete gravity dams to CDA guidelines.
- Timber framed residential design.
- Human induced vibrational analysis of lightweight structures.
- Insurance and building condition assessment.

Awards

- 32nd Sean DeCourcy Award, from the Irish Concrete institute
- 2019 ACEI Winner Category Structures Large (Team) - One Microsoft Place



Contact Information

matt.roche@ecora.ca

Education

BE (Hons) Structural Engineering, Dublin Institute of Technology – 2013

MSc., Structural Engineering & Mechanics, University of Edinburgh – 2014

Associations

- Engineers and Geoscientists British Columbia
- Institute of Structural Engineers (Graduate Member)
- Structural Engineers Association of British Columbia

Specializations

- FEA modelling and analysis
- Vibration serviceability analysis
- Dynamic earthquake analysis and design.
- Large and small scale Industrial, Commercial and Infrastructure Building design.

Years of Experience

9

Years with Firm

2-3

Office

Kelowna, BC

CONSULTANT PROFILE

Matthew Roche, P.Eng.
Structural Engineer

Relevant Experience

Infrastructure Sector

- **Chain Lake Dam Upgrades, Princeton, BC (2021 – Ongoing) –** Project involved the design of a new spillway for Chain Lake Dam. Structural EOR for the concrete sections of the spillway, including pedestrian bridge. Development of structural Autodesk Civil3D drawings.
- **Kitley Creek (Upper Prather) Dam Rehabilitation, Okanagan Falls, BC (2021 – Ongoing) –** Project was the design of a new dam, including concrete gravity sections, for St. Andrews by the Lake golf course. Structural EOR for concrete gravity sections of dam. Design and analysis of reinforced concrete sections, including the placement of reinforcing rebar. Development of structural Autodesk Civil3D drawings.
- **Frazer Lake Dam Upgrades, Kelowna, BC (2019 – 2022) –** Ecora was retained by the City of Kelowna to design dam upgrades for Frazer Lake Dam. Site inspection services during construction and post-construction. Development of structural Autodesk Civil3D drawings.
- **Rose Lake Dam Remediation, Williams Lake, BC (2020 – 2022) –** Project to upgrade Rose Lake Dam to protect downstream infrastructure. Independent reviewer of structural design and drawings.
- **Powell River Wastewater Treatment Facility, Powell River, BC (2019 – 2023) –** This $70 million project included the construction of three new buildings, including an administration building, large process mechanical building and the 6 exterior tanks in a Bioreactor building. Matthew was the lead design engineer for the process mechanical building which stored and treated the wastewater prior to transporting it for further treatment in the bio-reactor. The building was 2 storeys which were comprised of reinforced concrete walls, internal masonry walls, a concrete two-way spanning slab, and a steel deck roof.
- **Hells Gate Arch Wall Dam, Boston Bar, BC (2023) –** Performed full 3D finite element analyses for double curvature dam, including deflection calculations (see Image). Client was the Ministry of Transportation and Infrastructure.

Commercial and Industrial Sector

- **YVR Site Works, Underground Structures, Vancouver, BC –** Structural Engineer for the underground structures at Vancouver International Airport. Structural design of 7 m deep filter reinforced concrete filter chamber. Matthew designed many above and underground structures including external steel staircases, elevated generator pads and access ramps.

Insurance Sector

- **Muncho Lake Park Bridge Condition Assessment, Muncho Lake, BC –** Conducted a detailed structural assessment for a road bridge for the Ministry of Environment, Parks and Protected Area Division in Liard Area British Columbia. The bridge is a 71 m long, 1960's Bailey bridge. Recommended three different repair approaches with a corresponding service life and monitory cost for each.

CONSULTANT PROFILE

Patrick Machibroda, P.Eng.
Geotechnical Engineer

Summary of Experience

Patrick Machibroda is a geotechnical Engineer-in-Training (EIT) with over 7 years of experience in engineering practice across various Canadian provinces/territories including British Columbia, Alberta, Saskatchewan, Manitoba, Ontario, Nunavut, Yukon.

He has specialized knowledge of construction materials testing, high-end geotechnical testing (oedometer, direct shear, permeameter, tempe pressure cell, and triaxial) construction stage site reviews/inspections, 2D limit equilibrium and finite element modelling (slope stability, deformations/settlements, and seepage), deep foundation design, mechanically stabilized earth wall design, reinforced soil slope design, sub-surface investigations and in-situ testing.

Areas of Expertise / Specializations

- Construction Materials Testing
- Dam Embankment Stability Analysis
- High-end Geotechnical Laboratory Testing
- Slope Stability Analysis
- 2D Limit Equilibrium and Finite Element Modelling
- Sub-surface Investigation
- In-situ Testing
- Construction Site Reviews

Relevant Experience

- **Chain Lake Dam Upgrades, Princton, BC (2024) –** Stability analysis of embankment dam sections during various loading conditions. Wrote geotechnical portions of Plan Submission Report
- **Quintette (Shikano) Coal Mine Tailings Storage Facility, Tumbler Ridge, BC (2024 – Ongoing) –** Duties included developing preliminary 3D tailings deposition and breach analysis model using RIFT TD software.
- **Highway 1 Salmon Arm West (2024 – Ongoing) –** Performing consolidation and direct shear testing followed by interpretation of results for reporting and use.
- **BC Hydro Feeder Extension (2024 – Ongoing) –** Providing technical input on geotechnical investigations and analyses to a team of geotechnical engineers.
- **Metro Vancouver Coquitlam Water Main No. 4, Coquitlam, BC (2023) –** Role of geotechnical field engineer for a water main project. Conducted supplementary site investigations and assisted in coordination and reporting.



Contact Information
patrick.machibroda@ecora.ca

Office
Kelowna, BC

Education
University of Manitoba – M.Sc. in Civil Engineering – 2023

University of North Dakota – B.Sc. in Civil Engineering *cum laude* – 2021

College of Southern Idaho – A.A. in Liberal Arts *cum laude* – 2015

Associations / Memberships
Engineers and Geoscientists British Columbia (EGBC)

Years of Experience
+7

Years with Firm
< 1

Patrick Machibroda, P.Eng.
Geotechnical Engineer

- **Hodgson Slide, Williams Lake, BC (2023) –** Role of geotechnical modelling lead. Developed finite element slope stability, deformations, and settlements model post-landslide. Conducted sensitivity analysis of potential remediation efforts, communicated through PowerPoint presentation and technical memorandums. Performed direct client communication.
- **Canola Crush Facility, Regina, SK (2022 – 2023) –** Geotechnical Field Intern. Conducted preliminary site investigation, construction review of continuous flight auger piles, piezometric cone (CPTu) soundings, and analysis of instrumented static axial load test results.
- **Northern Hamlet of Patuanak Water Treatment Plant Upgrade, Patuanak, SK (2019 – 2020) –** Concrete Testing/Materials Technician. Conducted concrete testing, mechanical analysis, index testing of soils, and compaction testing by nuclear densometer.
- **Riverlanding East/North Tower, Saskatoon, SK (2016 – 2020) –** Concrete Testing/Materials Technician. Conducted concrete testing, mechanical analysis, and index testing of soils, and compaction testing by nuclear densometer.
- **Venus Tailings Facility Gap Analysis, Tagish, YK, (2023) –** Geotechnical Modelling Lead. Development finite element slope stability analysis model. Conducted sensitivity analysis of potential remediation efforts, communicated through PowerPoint presentation and technical report. Client: Crown-Indigenous Relations and Northern Affairs Canada (CIRNAC)
- **Long-term Care Centre, Rankin Inlet, NU, (2022) –** Geotechnical Engineering Intern. Conducted materials testing, gabion wall inspection, reporting, and compaction by nuclear densometer. Client: Nunavut Department of Health.

Publications

- “Estimating Pile Shaft Capacities Using Direct Methods for Continuous Flight Auger Piles in Silts and Sands”, Unive (Machibroda, 2023)

CONSULTANT PROFILE

Prajakta Jadhav, Ph.D. E.I.T,
Junior Geotechnical Engineer

Summary of Experience

Prajakta is a Geotechnical Consultant based in Ecora's Vancouver, BC office. Her background is in civil engineering with a specialization in geotechnical engineering with experience in static and seismic design of geotechnical structures and numerical modelling of soil-structure interactions.

She is proficient in a number of modelling and simulation software packages including: OpenQuake, OpenSees, ABAQUS, GEO-STUDIO, GEO-5 and MSEW.

Relevant Experience

- **Stabilization of silty ravines along Sabarmati River using Geosynthetics**: Performed comparative study of coir geotextiles with grouted rip-rap to inhibit surface erosion of silty ravines, design of gabion walls to impart global stability to the silty ravines.
- **Deformation-based seismic design of cantilever retaining walls:** Developed analytical double wedge model for predicting seismic sliding displacements of cantilever retaining walls simulating mechanism in backfill, design input for location of shear-key to improve the performance of retaining walls under seismic loading
- **Prediction of earthquake-induced lateral spread displacement using PSHA framework:** Performed PSHA in OpenQuake to estimate regional lateral-spread displacement curves for Canada, contributed GMPE source codes (Youd et a. 2002, Zhang & Zhao 2005) to OpenQuake GSIM library
- **Response of buried pipelines subjected to ground movements**: Estimated strain capacities of intact and corroded buried pipelines under ground movement, numerical modelling of pipeline subjected to longitudinal and transverse ground movements, studied the effect of varied ground displacement patterns on buried pipelines.

Publications & Conference Presentations

Jadhav, P.R, and Wijewickreme, D. (2022). "Regional liquefaction-induced ground displacement predictions for Canada Using 5th Generation NBCC seismic hazard model", *Canadian Geotechnical Journal, 59(*8), 1487-1502.

Jadhav, P.R. and Prashant, A. (2022). "Scaling factors quantifying seismic load uncertainty with soil nonlinearity effect in displacement-based design of bridge abutments." *Journal of Bridge Engineering, 27(*11), 04022110.

Jadhav, P.R. and Prashant, A. (2022). "Charts for permanent displacement-based seismic design of cantilever retaining walls", *Journal of Earthquake Engineering.* 2134943

Jadhav, P.R. and Prashant, A. (2020). "Computation of seismic translational and rotational displacements of cantilever retaining wall with shear key" *Soil Dynamics and Earthquake Engineering, 130,* 105966.



Office
Vancouver, BC

Education
Postdoctoral Research Fellow, University of British Columbia, Vancouver, BC

Ph.D. (Civil) Geotechnical, Indian Institute of Technology Gandhinagar, India

B.Tech (Civil Engineering), Veermata Jijabai Technological Institute, India

Associations / Memberships
Earthquake Engineering Research Institute
Canadian Geotechnical Society
Vancouver Geotechnical Society

Years of Experience
5+

Years with Firm
<1

CONSULTANT PROFILE

Melony Catana, AScT
Environmental Technologist

Summary of Experience

Melony Catana is a registered Applied Science Technologist that has been consulting in BC since 2005. She specializes in environmental management and permitting for infrastructure and land development projects. She is experienced with conducting terrestrial and aquatic wildlife and vegetation surveys, habitat assessments, impact assessments, project design, regulatory liaison, permitting and compliance monitoring. She has developed and implemented project-specific management plans for mitigating impacts to Species at Risk (SAR), including wildlife exclusion and salvages. She has extensive experience working as part multidisciplinary project teams for the remediation of dams, culverts, and bridges, including fish and wildlife salvages and monitoring watercourse diversions. Her design experience includes fish and wildlife habitat restoration and compensation. She has worked extensively in permit acquisitions and compliance under the Riparian Areas Protection Regulation (RAPR), the *Water Sustainability Act;* the *Fisheries Act* and municipal bylaws (development permits).

Relevant Experience

Fortis BC Interior Gas Upgrade (IGU) – As & When Services | 2023 to Present

- Provision of expert technical support and environmental auditing services for various gas line upgrade projects, including drilling investigations, restoration programs, and pipeline stream crossings. Work includes contributing to project planning and mitigation strategies to avoid or reduce risks to aquatic and sensitive habitats, regulatory liaison, and permit strategy development, and conducting fish habitat assessments.

City of Merritt – Flood Hazard Mitigation Options Analysis | 2023 to Present

- Conduced a biophysical site assessment; and is presently undertaking an options analysis, as part of a multi-disciplinary team of professionals, with inputs from other consultants, engineers, and community stakeholders, to examine options for dike and walkway reinstatements in consideration of environmental impacts, BMP, landowner agreements, public interest and permitting requirements.

BC Hydro – La Joie Dam Environmental Assessment | 2023 to Present

- Coordination of field work for an environmental assessment and environmental monitoring for geotechnical investigations on dam infrastructure, including overseeing and coordinating the field work program and safety training for Ecora and Tsal'alh First Nation personnel. Tsal'alh Development Corporation and Ecora are performing this work under their partnership at Szumin'ts.



Contact Information

melony.catana@ecora.ca

Education

Integrated Resource Management Technology, Sault College of Applied Arts and Technology, Sault Ste. Marie, ON

Ecosystem Management Technician, Sir Sanford Fleming College of Environmental and Natural Resource Sciences, Lindsay, ON

Associations

Applied Science Technologists & Technicians of BC (ASTTBC)

Specializations

- Environmental Assessment, Monitoring, Permitting & Liaison
- Fish & Wildlife Salvage
- Habitat Mitigation & Compensation Design
- Natural Resource Interpretation
- Project Management & Coordination

Years of Experience

18

Years with Firm

5

Office

Kelowna, BC

Melony Catana, AScT
Environmental Technologist

City of Vernon – 43rd Street Bridge Replacement & Flood Hazard Mitigation | 2023 to Present

- Conducted an environmental impact assessment, including a fish habitat assessment according to Fish Habitat Assessment Procedures (FHAP); prepared an environmental management plan, including habitat balance, restoration plans, compensation, cost estimates, tender documents and a long term effectiveness monitoring plan; and liaised with DFO and the Province for permit submissions.

City of Kelowna – Frazer Lake Dam Remediation | 2018 to Present

- Conducted field inventory surveys for wildlife species and vegetation; Assessed potential impacts to habitats and wildlife, including known and potential Species at Risk, based on three different dam remediation/water level scenarios; Worked within a multi-disciplinary project team, including engineers, technicians, specialist biologists and City staff, to develop engineering and mitigation designs that A) conformed with provincial and federal regulatory requirements and best management practices and B) met dam management objectives, while maintaining minimum habitat requirement for the Species at Risk present; Prepared an Environmental Management Plan including a habitat restoration/compensation plan. Provided support for provincial Dam Safety Regulation permitting and completed permitting under the *Wildlife Act* for wildlife salvages. Monitored construction, conducted wildlife salvage, completed salvage data reporting.

Chandos Construction – Grand Forks Flood Hazard Mitigation | 2022

- Performed environmental management, monitoring and reporting for integrated flood hazard mitigation works in the City of Grand Forks, which involved the construction of dikes, sheet piling, and horizontal directional drilling (HDD). Prepared Construction Environmental Management Plans (CEMP) based on existing permits (Section 11s and DFO authorizations) and EMP's prepared by the auditing consultant firm. This included producing management plans for Erosion & Sediment Control (ESC), spills and hazardous material, wildlife, Species at Risk, and emergency response for potential HDD fracking.
- Monitored water quality, inventoried vegetation, performed wildlife and breeding bird nest surveys. Developed management plans and 'No Disturb' buffers for active bird nests encountered, including those of the endangered Lewis' Woodpecker. Coordinated the development of plans during construction to address disposal of potentially hazardous HDD drilling fluids and potentially contaminated soils. Provided recommendations for risk management of environmental sensitivities encountered during construction. Provided daily environmental reports and communications. Coordinated and supervised junior personnel, coordinated senior contaminated sites specialist, and liaised with a large multidisciplinary project team.

CONSULTANT PROFILE

Adam Tieman, E.I.T., B.A.Sc
Junior Hydrotechnical Engineer

Summary of Experience

Adam Tieman is a Junior Hydrotechnical Engineer with our Hydrotechnical team supporting the group with design, modelling, drafting, and field reviews of projects in the Okanagan area. He is a 2020 graduate from the Civil Engineering program at UBCO. Adam has two years experience working at Ecora and has worked on projects performing drafting, modelling, design, and field services. Adam is experienced with a range of software including Autodesk Civil3D, HY-8, and FLO-2D. Adam's experience has been spread across B.C. with the majority of project in the Okanagan dealing with a variety of municipalities from RDOS to RDNO and MOTI projects.

Areas of Experience

- FLO-2D Dam Breach Modelling
- HY-8 Culvert Modelling
- Autodesk Civil3D Drafting & Surface Creation
- Civil Construction Field Reviews

Relevant Experience

Botanie Lake Dam DSR – 2023

- Assisting with OMS, DEP, & DSR report drafting

Becher & Toosey Dams DSR – Ongoing

- Modelling dam breach scenarios in FLO-2D
- Assisting with OMS, DEP, & DSR report drafting

Mamit Lake Dam DSR – 2023

- Modelling dam breach scenarios in FLO-2D
- Assisting with OMS, DEP, & DSR report drafting

Lodestone FSR Remediation – 2023

- Hydraulic modelling and culvert design
- Low volume road design



Contact Information

adam.tieman@ecora.ca

Education

University of British Columbia Okanagan – B.A.Sc. Civil Engineering – 2020

Associations

Engineers and Geoscientists of British Columbia (EGBC)

Years of Experience

2

Years with Firm

2

Office

Kelowna, BC

CONSULTANT PROFILE

Daniel Tamas, B.Sc.
Geoscientist in Training (GIT)

Summary of Experience

Daniel Tamas is a Geoscientist in Training with Ecora based in Vernon, BC. He is a graduate of the University of British Columba Okanagan with a Bachelor of Science B.Sc., in Earth and Environmental Sciences. He is registered in good standing with EGBC as a Geoscientist in Training (GIT) pursuing certification as a Professional Geoscientist.

He has 6 years of experience in consulting, 3.5 years in the environmental sector, and 2.5 years in the geotechnical sector. His geotechnical focus has been on coordinating and executing geotechnical drilling programs, developing terrain maps, and geotechnical assessment reporting. For field programs, he has performed the role of field technician and Prime Contractor for major highway projects and gas pipeline integrity digs. In the office, his focus is geotechnical reporting for completed field programs and desktop site assessments typically including factual geotechnical reporting and terrain mapping along existing and planned linear infrastructure.

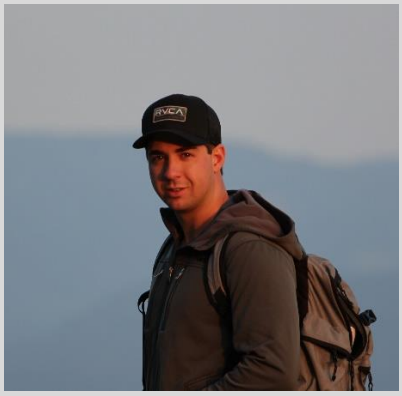
Areas of Experience

- Project coordination and implementation
- Safety coordination as Prime Contractor
- Scope and budget preparation and estimates/feasibility assessments
- Client, Contractor, and Owner communication and liaison
- Procurement of Contractor/Subcontractor and machinery support
- Quality Control / Quality Assurance
- Geotechnical report development
- Terrain and geohazard mapping
- Standard Penetration Test (SPT) Analyzer and data processing

Relevant Drilling Site Supervision Experience

Ministry of Transportation – As and When Contract: Highway 33 Dave's Creek Crossing, Chief Louis Way, Highway 3 Lamont Creek

- Performed the Prime Contractor duties leading site safety and construction coordination between utility locators, traffic control, drilling contractors, and first nations monitors.
- Identified and classified subsurface material using ODEX, Mud rotary, and Sonic drilling methods.
- Collected soil samples from drilling test holes, split spoon SPT, and test pits.
- Coordinated the use of SPT Analyzer, collected data, and processed data.
- Developed lab programs for soil sample testing.



Contact Information
daniel.tamas@ecora.ca

Education
Bachelor of Science in Earth and Environmental Science at the University of British Columbia Okanagan 2012—2017

Training and Certificates
- Occupational First Aid Level 1
- Workplace Hazardous Materials Information System (WHMIS)
- Radiation Safety & TDG – Portable Gauge Users
- CCiL – Concrete Testing Type QF certification

Years of Experience
6

Years with Firm
6

Office
Vernon, BC

Daniel Tamas, B.Sc.
Geoscientist in Training (GIT)

- Developed soil stratigraphy logs classifying encountered materials.

Fortis BC – Voght Creek Integrity Dig Intrusive Investigation

- Performed the Prime Contractor duties leading site safety and construction coordination between the client, their subcontractors, pipeline inspectors, and drilling contractors.
- Performed preliminary site assessments, including marking the site locates and identifying potential construction constraints such as loading requirements related to complex infrastructure designs.
- Identified and classified subsurface material using Sonic rigs to ensure proper depth was reached during exploration.
- Collected soil samples using boreholes, split spoon SPT, and test pits.
- Developed a geotechnical suitability assessment report detailing the pipeline repair options, including Horizontal Directional Drilling (HDD), auger, and trenched methods.
- Reviewed and assessed site conditions on steep slopes for temporary geohazard mitigation measures during integrity digs.

Relevant Geohazard and Terrain Mapping Experience

Ministry of Transportation and Infrastructure – Highway 1 Hoffmans Bluff

- Completed a historical imagery review and delineated terrain polygons based (TSIL E) based on the surficial material, surface expression, geomorphic qualifiers, and geomorphic processes.
- Developed a terrain characterization and sequence overview detailing the geomorphological history of the site.

Ministry of Transportation and Infrastructure – Highway 1 Remote Avalanche Control

- Completed a historical imagery review and delineated terrain polygons based (TSIL E) based on the surficial material, surface expression, geomorphic qualifiers, and geomorphic processes.
- Characterized an overview of terrain hazards detailing active and potential geomorphological processes which could impact infrastructure.
- Determined appropriate infrastructure placement to minimize geohazard interactions and maximize infrastructure lifespan.

CONSULTANT PROFILE

Abbas Rahman, PMP
Hydrotechnical Consultant

Summary of Experience

Abbas brings with him extensive experience as a civil design engineer and project manager. Starting his career as a project manager managing the reinstatement of damaged commercial buildings in the aftermath of the Christchurch earthquakes, Abbas progressed onto becoming a civil design engineer, designing and delivering infrastructure projects for municipalities, government agencies, businesses, and developers. Abbas' experience includes land development for residential and commercial subdivisions, designs for water, sanitary sewer and stormwater infrastructure, contract management, and assessment of drinking water infrastructure in accordance with Ministry of Health guidelines. Abbas has also led the preparation of tender documentation, evaluated bids, managed communications, and provided design changes during the construction phase.

Areas of Expertise / Specializations

- Hydrotechnical Assessment
- Project Management
- Municipal Engineering
- Land Development Engineering
- Site Servicing Design
- Tender Document Preparation
- Drainage Engineering

Relevant Experience

Maxwell Lake DSR, North Salt Spring Waterworks District (NSSWWD) (2024)

- Project Manager for the dam safety review of Maxwell Lake Dam on Salt Spring Island, liaising with NSSWWD staff, geotechnical team and principal consultants for the delivery of the Dam Safety Review (DSR).
- Undertook hydrotechnical analysis and inflow design flood (IDF) calculations.
- Drafted hydrotechnical section and dam risk assessment sections in the DSR.

Shikano North Tailings Facility Hydrological Analysis, Conuma (2024)

- Supported the Engineer of Record with due diligence study, assessing the condition of the dam, reviewing historical reports and deficiencies, and recommended remedial works to ensure compliance with the latest federal and provincial regulations.
- Undertook a comprehensive hydrological analysis including water balance assessment and derivation of an elevation-storage curve, catchment analysis, and Inflow Design Flood (IDF) and Probable Maximum Flood (PMF) calculations.

Rogers Creek FSR Deactivation, BC Ministry of Forest (2023)

- Undertook hydrological analysis, characterised debris floods/ flows scenarios and evaluated the effects of climate change to determine design flows within Rogers Creek for the removal of a culvert and deactivation of the Forest Service Road (FSR).
- Using HEC RAS to determine the optimal channel and riprap size, and alignment to safely convey flows during flood events, and collaborating closely with the geotechnical and geohazard teams, incorporating measures for bank stabilization.



Contact Information
abbas.rahman@ecora.ca

Office
Vancouver, BC

Education
BEng (Hons) – Natural Resources Engineering, University of Canterbury, New Zealand

Associations / Memberships
Chartered Professional Engineer (CPEng), Engineers New Zealand

Candidate for P.Eng. with Engineers and Geoscientists BC (EGBC)

Years of Experience
12+

Years with Firm
1

External Certificates
Project Management Professional (PMP)

Abbas Rahman, PMP
Hydrotechnical Consultant

214B Street Sanitary Lift Station Design, Township of Langley (2022 - Present)

- Civil Designer and Project Manager for the upgrades to a sanitary lift station in the Township of Langley. The wet well is to have a new, slip-in fibreglass liner, and the electrical kiosk, backup power supply generator and SCADA antenna are to be upgraded.
- Assessed options for specially made below ground chambers to house flow meter and overbuild manhole due to soft ground conditions coupled with high ground water levels.
- This project is currently in the construction phase, reviewing and recommending shop drawings for approval while working with the Contract Administrator for field reviews and payment claim approvals.

Becher and Toosey Dam OMS and DEPs, Tl'esqox of the Tsilhqot'in First Nation (2023)

- Drafted the OMS and DEPs for earthfill embankment dams which are situated along Riske Creek. The dams supply water to the First Nation community in Toosey IR No. 1. The tasks included gathering background information from various sources, historical design documents and drawings to identify the characteristics, operational and safety features of the dams, technical aspects such as the spillway levels, hydrological and hydraulic aspects of the dams including upstream and downstream features.
- Reviewing periodic surveillance, inspection and maintenance schedule, safety practices and infrastructure, emergency procedures, contingencies, water licenses and reporting procedures to ensure current procedures at the dam are in accordance with the Water Sustainability Ace and Dam Safety Regulation (BC Reg. 40/2016)
- Assisted in drafting the DSRs, reviewing hydraulic features, dam safety reporting activities and hydraulic features of the dam.

Botanie Lake Dam OMS, Lytton First Nation (2023)

- Drafted the OMS for the earthfill embankment dam situated along Botanie Creek. This dam provided water supply to the Lytton First Nation community.
- Reviewed periodic surveillance, inspection and maintenance schedule, safety practices, emergency procedures, water licenses, historical design documents and drawings to identify the characteristics, operational and safety features of the dams, and technical aspects such as the spillway levels, hydrological and hydraulic features.

Highway 3 Lamont Creek, BC MoTI (2022 — 2023)

- Hydrotechnical Designer for the replacement of a culvert under Highway 3 which was damaged during the 2021 atmospheric river flooding event. Several options were assessed for the replacement of the culvert. Some of the challenges included hydraulic and hydrology assessment for the large catchment with limited hydrometric data in close proximity and depth of culvert pipe cover under the highway.

BC-750 Culvert Assessment, Michels Canada, BC, (2022)

- Hydrotechnical assessment and design for a culvert along a fish-bearing creek where Michel Canada's pipeline crosses. The culvert outlet support had been undermined due to erosion over time.
- The capacity of the existing culvert was assessed in accordance with MoTI Supplementary to TAC Geometric Design Guide and was found to be inadequate. So, a design for a new upgraded culvert in accordance with Fish-stream Crossing Guidebook by Fisheries and Oceans Canada was provided, including riprap rock armouring to prevent scouring at the outlet.

Anderson Creek Bridge Assessment, Michels Canada, BC (2022)

- Hydrotechnical Designer to verify the elevation of a proposed temporary bridge which is to be constructed over Anderson Creek to support the construction of the Trans Mountain Pipeline.
- The scope included site investigations, catchment analysis, and hydrological and hydraulic assessment to determine the water elevation in the creek during a flood event. Given the catchment characteristic, effects of debris flood and debris flood had to be considered too.

Highway (SH1) Runoff Treatment, Te Awa Lakes, NZ (2022)

- Lead designer for stormwater swale channels to treat and convey road storm runoff from State Highway 1 in Waikato. The channel was sized to provide adequate capacity and treatment for the runoff.

Abbas Rahman, PMP
Hydrotechnical Consultant

- Channel outlet designed to discharge treated runoff to the Waikato River. Since the discharge point is submerged during a 1% annual exceedance probability storm event, a tide gate was required to prevent backflow into the swale channel. The swale channel was also designed large enough to detain the runoff during such a storm event.

Stormwater Network Contamination, KiwiRail, NZ-wide (2019 to 2020)

- Investigated source of contamination for storm sewer networks that discharge to the environment.
- Designed network upgrades for sanitary sewer, stormwater, and oil waste networks for sites across the North Island.
- Assisted the Capital Programme Manager with programme planning for infrastructure renewal.
- Responsible for the design, tender evaluation, and construction administration for sanitary sewer and trade waste upgrades at the Auckland and Hamilton train depot.

CONSULTANT PROFILE

Bram Samuels, M.Eng., E.I.T.

Junior Geotechnical & Hydrotechnical Engineer

Summary of Experience

Bram Samuels is a Junior Geotechnical & Hydrotechnical Engineer in Ecora's Vancouver office. Bram has four years of broad experience in geotechnical, hydrogeological, hydrotechnical, and coastal engineering site investigation, analysis, and design, across the Lower Mainland and Vancouver Island, in highly variable soil, groundwater, and surface water conditions.

Bram has completed engineering consulting work for a variety of client types, including other engineers, earthworks contractors, municipalities, small and large residential and commercial developers, the Ministry of Transportation (MoTI), and First Nations groups.

For each of his projects Bram has been involved from the investigation phase all the way through to the design and construction phases and has been in technical communication with every level of employee from field workers/EITs to superintendents/managers to CEOs. He therefore understands how to tailor technical communications to each audience.

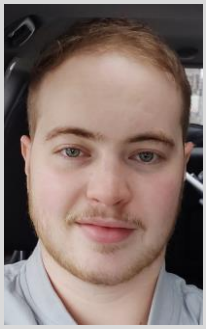
Bram is a strong project manager and field engineer with experience managing, coordinating, and conducting site investigation programs and field reviews with a small team of younger EITs and technical laboratory staff (including himself).

Bram has successfully conducted more than 35 site investigations in all types of soil and rock across the Lower Mainland and Vancouver Island in the last four years, including the design and installation of monitoring wells and piezometers. These investigations were completed using auger, sonic, ODEX, or Mud Rotary drills.

Bram is familiar with the requirements of the 2019 Vancouver Building Bylaw (VBBL), 2020 National Building Code of Canada (NBCC), 2019 Canadian Highway and Bridge Design Code (CHBDC) S6-19, BC MoTI Supplement to CHDBC S6-19, and the 2020 BC MoTI Standard Specifications as these documents pertain to shallow and deep foundations, embankment slopes, hydrologic analysis, and hydraulic structures.

Bram can complete analyses and design for excavation shoring systems, soil and rock slopes, shallow and deep foundations, ground improvement systems, retaining walls, various piping systems, spillways, culverts, and coastal engineering systems.

Bram is experienced with specialized geotechnical, hydrogeological, hydrotechnical, and coastal engineering modelling software (and conventional coding languages with useful software packages) such as Geostudio Suite (Slope/W, Seep/W, Sigma/W, Quake/W), L-Pile/RS-Pile, Settle3, ProShake, GRL-WEAP, SupportIT/SpanIT, LiqSVs, SRwall, Gwall, WallAP, MODFLOW, TopoDrive/ ParticeFlow, HEC-RAS, HEC-HMS, HY-8 and FLO-2D, QGIS, Python, and Matlab.



Contact Information

bram.samuels@ecora.ca

Education

M.Eng. in Civil Engineering
University of British Columbia - 2022

B.A.Sc. in Geological Engineering
Queen's University - 2018

Associations

Association of Professional Engineers & Geoscientists BC (APEGBC)

Vancouver Geotechnical Society (VGS)

The Canadian Society for Civil Engineers (CSCE)

Training

First Aid-Emergency First Aid for Industry (OFA Level 1), Nov. 2020

WHMIS 2015

Indigenous Awareness 101 & 201

Years of Experience

4

Years with Firm

1

Office

Vancouver, BC

Bram Samuels, M.Eng., E.I.T.

Junior Geotechnical & Hydrotechnical Engineer

Relevant Experience

- **Pier West Excavation Shoring (Field Reviews) –** Main field EIT at a 10-15 m deep excavation adjacent to the Fraser River and three major rail lines. Reviewed jet grout wall installation. Reviewed secant wall shoring using backflow preventers adjacent to major rail lines and dealt with related sinkholes. Reviewing pipe pile installations and completing static load tests for these piles. Subgrade reviews. Working with many contractors from different trades.
- **YVR Airport - CORE Program** – This site has extremely high consequence factors and is built on liquefiable sands that required extensive ground improvement. Reviewed stone column and RAP installations for parkade and utility building. Completed many stone column load tests. Review of slopes related to long-term shutdown of program. Vibration monitoring of Canada Line piers. Subgrade reviews. Working with many contractors from different trades.
- **YVR Pedestrian Facilities** – Design of helical piles to support the rain protection structures over the pedestrian pickup area outside of YVR. Design of helical piles and field reviews of installations for new YVR warehouses along Miller Rd.
- **60 Isleview Place, Lions Bay, BC** – Design of retaining walls, stormwater detention/infiltration tank, and drainage system for new construction on very steep slope. Field reviews and design change memos to bring project to fruition.
- **Fairbanks Load Landslide Remediation, 1108/1110 Fairbanks Road, Cowichan Bay, BC** – Design of a pipe system to convey flows down a cliff where previous overland flooding caused a surficial slope failure. This design includes slope stabilization/armouring, inlet and outlet structure design, pipe anchor design, and coastal engineering analysis.
- **Nunns Creek Culvert Replacement, Highway 19A ~150 m West of Spit Road, Campbell River, BC** – Site investigation and pile design for culvert replacement. This included an auger test hole, mud rotary test hole, sonic test hole, CPT/sCPT sounding, and downhole compressional and shear wave velocity sounding. Also included liquefaction analyses, constructability considerations, discussions on affecting sewer with the City of Campbell River.
- **Lefferson Creek Retaining Wall Design, 5260 Goldspring Place, Chilliwack, BC** – Replacing an existing timber retaining wall with a new wall comprised of lock blocks and geogrid. Significant utility conflicts and constraints.
- **Glenmore Voltage Conversion Project (Overpass Loading), 15th St and Highway 1, West Vancouver, BC** – Slope stability modelling looking at suitability of open trenching below footings for major Highway 1 overpass bridge. Also included subgrade reviews of vaults/manholes for other sections.
- **Seton Portage Debris Flow Mitigation Project, Seton Portage, BC** – Mitigation system design for debris flow hazard from Goat Mountain.
- **Seton Portage Engineering Services to Support Hotel, Seton Portage, BC** – Hydrological and hydraulic analyses to determine flood control elevation for the hotel, as well as general geotechnical engineering services.

CONSULTANT PROFILE

Meghan Sherwood, E.I.T.

Junior Geotechnical Engineer

Summary of Experience

Meghan Sherwood is a Junior Geotechnical Engineer based in Ecora's Penticton, BC office. She has experience conducting ARD/ML assessments and site reconnaissance, geotechnical site investigations, terrain analysis and mapping, slope stability analysis, retaining wall design, lab testing and construction monitoring. She is proficient with industry standard geotechnical slope stability and seepage modelling software such as GeoStudio Suite, Rocscience (Slide2, RocPlane, RocFall, SWedge, Dips) and gINT.

Areas of Expertise / Specializations

- Conducting Acid Rock Drainage and Metal Leaching (ARD/ML) and riprap suitability assessments for numerous quarries and rock sources in BC.
- Geotechnical site investigations and assessments.
- Slope stability assessments, modelling, and remediation recommendations.
- Construction monitoring and field reviews, including on-site recommendations for highway construction, riprap slope armouring and residential construction.

Relevant Experience

FSR Riprap Source Assessments, BC – BC Ministry of Forests (2022-present)

- Preliminary and detailed ARD/ML assessments for numerous potential riprap sources along Forest Service Roads, comprising site reconnaissance and bedrock sampling, laboratory analysis, geochemical characterization of rock samples, and assessing the potential for the rock source to generate metal leaching and acid rock drainage. Preparing ARD/ML prediction and mitigation plans.

Ntequem East Quarry ML/ARD Assessments, Ashcroft, BC (2023-2024)

- Detailed ML/ARD assessment for existing rock quarry comprising site reconnaissance and bedrock sampling, laboratory analysis, geochemical characterization of rock samples, and assessing the potential for the rock source to generate metal leaching and acid rock drainage.

Genelle Quarry ML/ARD Prediction, Genelle, BC (2023)

- Detailed ML/ARD assessment for existing rock quarry and aggregate pit and preparing a ML/ARD Prediction and Mitigation Plan.

2021 Emergency Flood Response Works, Hwy 1, 3, 5, 8 – BC MoTI (2021—2023)

- Preliminary and detailed ARD/ML and riprap suitability assessments for numerous potential rock sources utilized in emergency response and recovery repairs due to the 2021 atmospheric river flooding event.

Hwy 3 Rock Source, Princeton, BC (2022)

- Detailed ARD/ML and riprap suitability assessments for proposed rock source on private property to provide riprap material for mitigation measures of a roadway due to a slope failure caused by river erosion.



Contact Information
meghan.sherwood@ecora.ca

Office
Penticton, BC

Education
B.A.Sc. in Geological Engineering, University of British Columbia

Associations / Memberships
Member Engineers and Geoscientists British Columbia (EGBC)

Years of Experience
3

Years with Firm
2+

Certifications

- Metal Leaching/Acid Rock Drainage Field School – Bulkley Valley Research Centre, 2023.
- Introduction to Mineral Exploration Safety Course - Association for Mineral Exploration, 2020.
- Occupational First Aid – Level 1
- Ground Disturbance for Workers
- Fall Protection Worksite Safety
- Radiation Safety & TDG – Portable Gauge Users

CONSULTANT PROFILE

Tomos Edmonds, E.I.T.

Junior Hydrotechnical Engineer

Summary of Experience

Tomos Edmonds is a junior hydrotechnical engineer currently working with Ecora Engineering & Environmental Ltd. He has worked with Ecora for 5 years throughout the summer months working on a wide range of projects within the civil, geotechnical and hydrotechnical engineering disciplines.

Tomos is proficient in AutoCAD Civil3D and is working towards specializing in hydrotechnical analysis, modelling, and design, with a special interest in emergency management within the hydrotechnical sector.

Areas of Expertise / Specializations

- 2D/3D CAD drafting and modelling in Civil, Geotechnical & Hydrotechnical engineering disciplines
- Hydrotechnical engineering design including hydrological modeling
- Dam Safety
- 3D Stability Modeling of Concrete Gravity Dams (CADAM3D)

Relevant Experience

City of Vernon - Vernon Creek Rehabilitation

- Drafting Assistant to Vernon Creek rehabilitation in Vernon, BC. Tasks included 2D hydraulic modeling of the subject reach of Vernon Creek, and Civil3D Construction drawing design and production.

Town of Ladysmith – Holland Creek Dam Safety Inspections & Decommissioning

- Dam safety inspection reporting for three dams owned and operated by the Town of Ladysmith. Decommissioning plan for one of the dams including Civil3D construction drawing design and production.

BC Hydro Bridge River 1 Generating Facility – Slope Stabilization & Remediation

- Slope stabilization and rehabilitation construction drawing design and production since May 2022.



Contact Information
tomos.edmonds@ecora.ca

Office
Kelowna, BC

Education
BaSC – Civil Engineering

Associations / Memberships
Engineer in Training (E.I.T) - Engineers and Geoscientists of British Columbia (EGBC)

Years of Experience
1.5

Years with Firm
5

Appendix B

Project Profiles

PROJECT PROFILE

Yellow Lake Dam Upgrades



PROJECT DESCRIPTION

An engineering assessment of the Yellow Lake Dam was completed in 2017 by Ecora. This assessment indicated that the dam was not in compliance with the Provincial Dam Safety regulations or the Canadian Dam Safety Association Review Guidelines. The assessment report recommended that several action items be met to bring the dam into compliance. The recommendations included:

- Complete works to ensure the dam has freeboard to handle wind and wave effects and is capable of handling the Inflow Design Flood (IDF);
- Remove blockages of the low-level outlet and ensure proper function of the outflow pipe and gate; and
- Increase channel capacity, to prevent overtopping of banks along Highway 97. Reduce the potential for highway overflow and erosion.

Based on the findings of the engineering assessment, in 2021 Ecora completed engineering and environmental services to prepare and submit construction ready engineered designs and acquire all necessary authorizations for the required dam structural and operational upgrades as described above. This work also included the preparation of a detailed tender ready documents, Class 1 cost estimates and corresponding costing reports, contract administration, inspection services, environmental monitoring during the construction, production of record drawings and delivery of a final construction report.

Client

Ministry of Forests, Lands, Natural Resource Operations and Rural Development

Location

Penticton, BC

Duration

July 2016 — October 2021

Project Team

Michael J. Laws, P.Eng.
Adrian Chantler, Ph.D., P.Eng.
Jeff Redwood, P.Eng.
Pete Wittstock, P.Eng.
Adam Kerk-hecker, P.Eng.
Andrew Gain E.I.T.

Key Features

- Background review and site reconnaissance of the dams
- Geotechnical investigation.
- Geotechnical assessment including, seepage, static and seismic embankment stability, piping failure and liquefaction.
- Determination of Inflow Design Flood (IDF) and hydraulic assessment of spillway structures.
- Engineered conceptual design of remediation options for the dam structure and spillway.
- Completion of design reports and budgetary construction cost estimates.
- Preliminary and detailed design for the preferred option.
- Tendering and construction services.

Fees

$114,000

Client Contact

Tara White, R.P.Bio
778.622.6839
Tara.white@gov.bc.ca

PROJECT PROFILE

Cannell Lake Dam Remediation



PROJECT DESCRIPTION

In 2018, Ecora conducted an engineering assessment and conceptual remediation design for the Cannell Lake earthfill dam owned and operated by the City of Abbotsford. The assessment was performed in response to deficiencies identified in the most recent Dam Safety Review (Ecora, 2016).

Following the engineering assessment, Ecora proceeded with engineering designs and prepared a design report, recommendations, and budgetary construction cost estimates for the Works. Ecora was further retained to prepare a detailed design package for the preferred option and to tender the project. Contract administration and construction supervision was also completed as part of the project.

Ecora conducted background reviews, topographical surveys, geotechnical investigations and materials testing, hydrotechnical analyses that included determining the Inflow Design Flood's (IDF), hydraulic assessments of the spillway structures including 1D and 2D modelling (HECRAS) and undertook a series of geotechnical assessments for potential embankment failure modes, including embankment overtopping, piping through the embankment, piping through the foundation, static and seismic downstream and upstream stability and liquefaction. The analysis included the review of previous design information.

Ecora completed all aspects of the project in accordance with the requirements of the BC Dam Safety Regulation (BC Reg. 44/2016) and the Canadian Dam Association 2007 Dam Safety Guidelines (2013 Edition).

Client

City of Abbotsford

Location

Mission, BC

Duration

July 2018 — December 2020

Project Team

Michael J. Laws, P.Eng.

Adrian Chantler, Ph.D., P.Eng.

Chelsea Evans, B.E. (Hons) Civil

Pete Wittstock, P.Eng.

Adam Kerk-hecker, P.Eng.

Andrew Gain E.I.T.

Key Features

- Background review and site reconnaissance of the dams
- Geotechnical investigation.
- Geotechnical assessment including, seepage, static and seismic embankment stability, piping failure and liquefaction.
- Determination of Inflow Design Flood (IDF) and hydraulic assessment of spillway structures.
- Engineered conceptual design of remediation options for the dam structure and spillway.
- Completion of design reports and budgetary construction cost estimates.
- Preliminary and detailed design for the preferred option.
- Tendering and construction services.

Fees

$196,000

Client Contact

Kristi Alexander, P.Eng.

604.864.5502

kalexander@abbotsford.ca

PROJECT PROFILE

Centre Star Gulch Reservoir Dam Safety Review



PROJECT DESCRIPTION

Ecora in partnership with ISL Engineering and Land Services Ltd. conducted the 2015 Comprehensive Dam Safety Review of the Centre Star Gulch Reservoir for the City of Rossland, BC. The reservoir is formed through the construction of two dams within a north south trending valley providing an upland source of potable water for the City of Rossland, BC. The maximum height of the two dams vary from 4.1 m to 4.6 m and are each between 55 m and 118 m long.

Ecora completed all aspects of the Dam Safety Review in accordance with the requirements of the BC Ministry of Environment Dam Safety Guidelines (2012) and the Canadian Dam Association 2007 Dam Safety Guidelines (2013 Edition).

Ecora conducted a back ground review, site reconnaissance, failure mode assessment, dam breach analysis, consequence classification review of the two dams, reviewed the inundation mapping and hydraulic assessment of the spillway structure, and undertook a series of geotechnical assessments for potential embankment failure modes, including embankment overtopping, piping through the embankment, piping through the foundation, static and seismic downstream and upstream stability and liquefaction.

The geotechnical assessment also included a seepage analysis to determine the input ground water conditions in the embankments for the stability assessment.

A review of the dam's system management system and a dam safety expectations assessment utilizing the BC MFLNRO Dam Safety Branch, Dam Safety Expectations check sheet was also undertaken.

CLIENT

City of Rossland

LOCATION

Rossland, BC

DURATION

June 2015 to October 2015

PROJECT TEAM

Michael J. Laws, P.Eng.

Pete Wittstock, E.I.T.

KEY FEATURES

- Dam Safety Review in accordance with the requirements of the British Columbia Water Act, the BC Ministry of Environment Dam Safety Guidelines (2012), the BC Dam Safety Regulation and Canadian Dam Association 2007 Dam Safety Guidelines (2013 Edition).
- Background review and site reconnaissance of the dam
- Failure mode assessment.
- Dam break analysis and consequential classification review.
- Geotechnical assessment including, seepage, static and seismic embankment stability, piping failure and liquefaction.
- Review of dam safety management system.
- Dam safety expectation assessment.

Contract Amount

$15,950

Contract Type

Lump Sum Fixed Fee

PROJECT PROFILE

Shane Lake Dam Safety Review, Dam Remediation, and Dam Safety Review Updates



Client

The City of Prince George

Location

Prince George, BC

Duration

March 2020 — Ongoing

Project Team

Michael J. Laws, P.Eng.
Adrian Chantler, Ph.D., P.Eng.
Adam Kerk-Hecker, P.Eng.
Glen McCrae, P.L.Eng.
Adam Patterson, R.P.Bio

Key Features

- Dam safety review.
- Background review.
- Stakeholder engagement.
- Site investigation.
- Hydrotechnical analysis for upgrade design.
- Detailed design package preparation.
- Environmental assessment.
- Completion of design reports and construction cost estimates.
- Tender package preparation.
- Contract administration.
- Construction and post construction services.

PROJECT DESCRIPTION

Shane Lake Dam is a zoned earthen embankment dam with the function of providing redundancy to existing beaver dams at the south end of Shane Lake in the event one of the beaver dams fail. The dam is maintained and operated by the City of Prince George (the City) and has been assigned a consequences classification of “Very High”, as defined in the BC Dam Safety Regulations (BC. Reg. 40/2016) and the 2007 Canadian Dam Association (CDA) Dam Safety Guidelines (2013 Edition).

Ecora conducted a Dam Safety Review (DSR) for the structure in 2020, and based on the findings, Ecora was asked to provide engineering and environmental services related to remediation design and construction. The required improvements included:

- Improvements to the access road that leads to the dam; and.
- Replacement of the existing piped overflow system with a new auxiliary open channel spillway.

The open channel spillway was cut into the right abutment of the dam embankment and featured a control section constructed of articulated concrete block mats and 70 meters of riprap lined channel with a berm constructed for the left bank. The channel then discharged into a fish-bearing stream (Shane Creek). To facilitate these improvements to the dam, Ecora provided the City with the engineering and environmental services necessary to complete site investigation, detailed design and implementation plan, tender preparation and support, construction services and post construction services.

PROJECT PROFILE

Frazer Lake Dam Rehabilitation



CLIENT

City of Kelowna

LOCATION

Kelowna, BC

DURATION

July 2019 to Nov 2022

PROJECT TEAM

Michael J. Laws, P.Eng.
Adrian Chantler, Ph.D., P.Eng
Adam Kerk-Hecker, E.I.T.
Andrew Gain, E.I.T.
Glen McCrae, P.L.Eng.
Jordan Bokla, Int. Tech.
Adam Patterson, R.P.Bio., QEP
Melony Catana, AScT, Env. Tech.

PROJECT DESCRIPTION

Ecora provided engineering services to the City of Kelowna for the rehabilitation of Frazer Lake Dam, located in the Upper Mission neighbourhood of Kelowna, BC.

The preliminary design included review of existing documentation, conducting site reconnaissance including a geotechnical subsurface investigation, geotechnical analyses, hydrotechnical analysis, development of design alternatives, evaluation of design alternatives and a class C cost estimate. At the completion of the preliminary design the project proceeded to detailed design and completion of tender ready IFC drawings and schedule of quantities.

Environmental impact assessments were completed by the Ecora team which included several surveys of wildlife in the area. A construction environmental management plan was completed as part of the BC Dam Safety Regulation Plan Submission Requirements.

Details of the dam remediation design included:

- Removal of existing earthen embankment, concrete headwall, and outlet works;
- Temporary cofferdam design;
- Steel sheet pile cut-off wall;
- Rock ramp spillway channel;
- Cast-in-place concrete level control structure; and
- Low-level outlet and gate works;

KEY FEATURES

- Site reconnaissance including geotechnical subsurface investigation.
- Hydrotechnical analysis including rainfall analysis, dam breach analysis and flood routing.
- Environmental services including wildlife survey, impact assessment, and CEMP.
- Development of rehabilitation alternatives up to and including a class C cost estimate and evaluating benefits of each option.
- Detailed design of preferred option including a Class A cost estimate.
- Completion of Plan Submission Requirements for MFLNRORD approval.
- Procurement and Contract Administration services
- Construction and Environmental Monitoring services

Frazer Lake Dam Rehabilitation



FEES
$140,000

CONSTRUCTION VALUE
$761,000

CLIENT CONTACT
Robinson Puche
250.469.8480
rpuche@kelowna.ca

Brian Beach
250.469.8726
bbeach@kelowna.ca

This project had challenging construction issues with environmental requirements for preservation of amphibians, terrestrials, and groundwater seepage within the existing dam. Ecora prepared Request for Proposal documents for construction and participated in the selection process for the successful contractor. The project was constructed in 2022. The contractor chose to install a temporary sheet pile dam to isolate the dam removal and reconstruction works. Ecora is provided construction environmental monitoring services including nesting surveys, aquatic and terrestrial sweeps, contract administration, engineering oversight and field reviews.